

The University of Faisalabad

LAB MANUAL

Course Title: Data Mining

Submitted To:

Mr. Arham

Submitted By:

Mohammad Mukedas

2023-BS-AI-008

Department of Artificial Intelligence

Table of Contents

Lab 1: KNIME Introduction.....	4
Lab 2: Customer Churn Prediction.....	12
Lab 3: Data Cleaning.....	18
Lab 4: Data Preprocessing.....	23
Lab 5: Market Basket Analysis.....	30
Lab 6: Apriori Algorithm (Product Combination Analysis).....	35
Lab 7: FP-Growth (Efficient Frequent Pattern Mining).....	40
Lab 8: Python Basics (Reusable Data Processing Module).....	45
Lab 9: Decision Tree.....	50
Lab 10: Naïve Bayes & KNN.....	56

Lab 11: SVM.....	61
Lab 12: Clustering.....	67
Lab 13: Outlier Detection.....	72
Lab 14: Scraping and Sentiment Analysis.....	74
Lab 15: Final Capstone.....	76

Lab 1: KNIME Introduction

Real-World Scenario

A retail company stores large volumes of transaction records from different product categories. Although the data is available, it cannot support decision-making unless it is organized and examined properly. Business managers require clear analytical outputs to measure overall sales revenue, estimate average customer purchase value, identify the categories generating the greatest earnings, and determine which individual products produce the highest profit contribution.

This practical task uses **KNIME Analytics Platform** to transform raw retail sales data into useful business intelligence through a workflow-based analytical process.

Dataset Description

The dataset applied in this exercise is a **Retail Sales CSV file** containing detailed transaction records.

Each row represents one completed sale and includes fields such as:

- Transaction ID
- Product Title
- Product Category
- Quantity Sold
- Unit Price
- Purchase Date
- Sales Amount (or Revenue generated during processing)

The dataset is suitable for performing summarization, revenue calculations, category comparisons, and product ranking analysis.

Software Tool Used

KNIME Analytics Platform

KNIME is a visual analytics software that allows users to create data workflows by connecting functional nodes. Every node is responsible for a specific task such as importing files, cleaning data, changing data types, calculating formulas, generating summaries, or performing aggregations.

When combined, these nodes build a complete data analysis pipeline that converts unstructured sales records into valuable managerial insights.

Practical Learning Outcomes

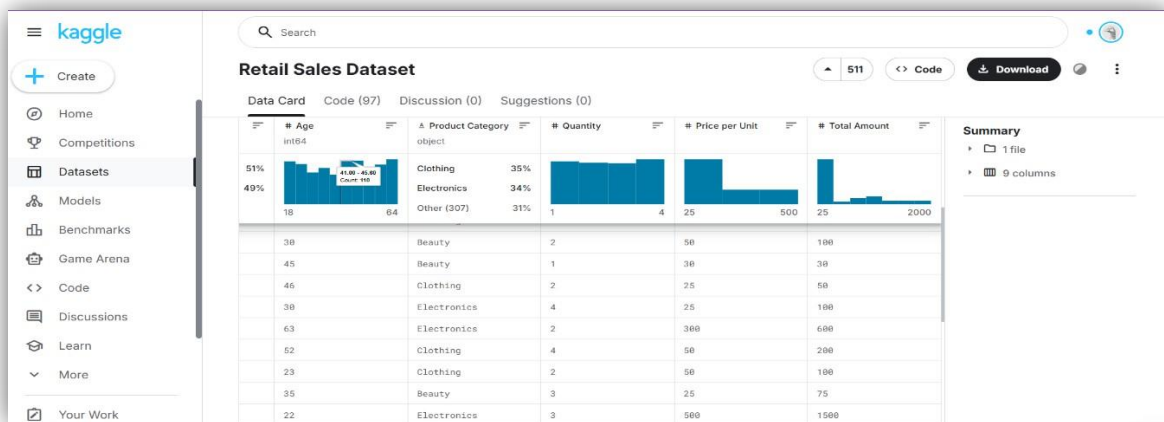
After completing this lab activity, students will be able to:

1. Load a CSV sales dataset into KNIME using the **CSV Reader** node.
2. Inspect and adjust column data types for accurate processing.
3. Produce descriptive statistics to understand the nature of the dataset.
4. Create a new **Revenue** field using the **Math Formula** node.
5. Analyze four major retail KPIs using **GroupBy** and **Sorter** nodes.

Procedure

Step 1: Download the Dataset

Obtain the Retail Sales dataset in CSV format from the provided source and save it locally for analysis.



Step 2: Create a New Workflow

Launch KNIME Analytics Platform and create a new workflow named *Retail Sales Analysis* to organize the analytical process.

The screenshot shows the 'Create a new workflow' dialog box in the KNIME Analytics Platform. The dialog box has a title bar with a close button. Inside, there is a label 'Workflow name' and a text input field containing the text 'Retail Sales Analysis'. At the bottom right, there are two buttons: 'Cancel' and 'Create'.

Step 3: Import the Dataset

Use the CSV Reader node to load the Retail Sales dataset into the workflow environment and verify successful execution.

File

Settings Transformation Advanced Settings Limit Rows Encoding Flow Variables Job Manager Selection Memory Policy

Input location

Read from: Local File System

Mode: ☒ File ☐ Files in folder

File: C:\Users\Qadri Laptop\Downloads\archive\retail_sales_dataset.csv

Reader options

Format

Autodetect format

Column delimiter: ,

Row delimiter: ☒ Line break ☐ Custom

Quote char: "

Quote escape char: \

Comment char: #

Has column header: ☒ Has RowID: ☐ Support short data rows: ☐ Prepend file index to RowID: ☐

Preview

The suggested column types are based on the first 10000 rows only. See 'Advanced Settings' tab.

Row ID	Transac...	Date	Custom...	Gender	Age	Product ...	Quantity	Price pe...	Total A...
Row0	1	2023-11-24	CUS T001	Male	34	Beauty	3	50	150
Row1	2	2023-02-27	CUS T002	Female	26	Clothing	2	500	1000
Row2	3	2023-01-13	CUS T003	Male	50	Electronics	1	30	30
Row3	4	2023-05-21	CUS T004	Male	37	Clothing	1	500	500
Row4	5	2023-05-06	CUS T005	Male	30	Beauty	2	50	100
Row5	6	2023-04-25	CUS T006	Female	45	Beauty	1	30	30
Row6	7	2023-03-13	CUS T007	Male	46	Clothing	2	25	50
Row7	8	2023-02-22	CUS T008	Male	30	Electronics	4	25	100
Row8	9	2023-12-13	CUS T009	Male	63	Electronics	2	300	600
Row9	10	2023-10-07	CUS T010	Female	52	Clothing	4	50	200
Row10	11	2023-02-14	CUS T011	Male	23	Clothing	2	50	100

OK Apply Cancel ?

Step 4: Verify the Data Types

Inspect all columns to ensure numeric fields (Quantity, Price, Revenue) are correctly typed as Integer or Double, and categorical fields (Category, Product Name) are set as String.

► 1: File Table

Rows: 1000 | Columns: 9

Table

#	RowID	Transacti...	Date	Customer ...	Gender	Age	Product C...	Quantity	Price per ...	Total Amo...
1	Row0	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
2	Row1	2	2023-02-27	CUST002	Female	26	Clothing	2	500	1000
3	Row2	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
4	Row3	4	2023-05-21	CUST004	Male	37	Clothing	1	500	500
5	Row4	5	2023-05-06	CUST005	Male	30	Beauty	2	50	100
6	Row5	6	2023-04-25	CUST006	Female	45	Beauty	1	30	30

Step 5: Generate Summary Statistics

Apply the Statistics node to compute descriptive measures such as mean, minimum, maximum, sum, and standard deviation for numeric attributes.

The screenshot shows the Alteryx workflow with a CSV Reader node connected to a Statistics node. The Statistics node is configured with the 'Manual' column filter. The 'Includes' list contains Transaction ID, Date, Customer ID, Gender, and Age. The 'Excludes' list is empty. The 'Apply and Execute' button is visible.

Below the workflow, the '1: Statistics Table' is displayed. It shows 5 rows and 16 columns. The table contains summary statistics for the following columns:

#	RowID	Column	Min	Max	Mean	Std. devia...	Variance	Skewness	Kurtosis	Overall su...	No. r
1	Transa	Transaction ID	1	1,000	500.5	288.819	83,416.667	0	-1.2	500,500	0
2	Age	Age	18	64	41.392	13.681	187.182	-0.049	-1.201	41,392	0
3	Quantit	Quantity	1	4	2.514	1.133	1.283	-0.014	-1.393	2,514	0
4	Price p	Price per Unit	25	500	179.89	189.681	35,979.017	0.736	-1.139	179,890	0
5	Total A	Total Amount	25	2,000	456	559.998	313,597.347	1.376	0.815	456,000	0

Step 6: Calculate Revenue Column

Use the Math Formula node to create a new Revenue column using the formula:

$$\text{Revenue} = \text{Quantity} \times \text{Price}$$

This derived column will be used for KPI calculations.

The screenshot shows the 'Dialog - 3:3 - Math Formula' window. The 'Math Expression' tab is selected. The 'Column List' on the left contains Transaction ID, Age, Quantity, Price per Unit, and Total Amount. The 'Function' list on the right includes ROWCOUNT, ROWINDEX, pi, COL_MIN, COL_MAX, COL_MEAN, COL_MEDIAN, COL_SUM, COL_STDEV, COL_VAR, and log. The 'Expression' field contains the formula: `$Quantity * $Price per Unit`. The 'Append Column' option is selected, and the new column name is 'new column'.

▶ 1: Group table		Flow Variables	
Rows: 1		Columns: 1	
☐	#	RowID	Sum(Quantity) Number (Integer)
☐	1	Row0	2514

Step 7: Calculate Total Revenue (KPI 1 – Total Revenue)

Add a GroupBy node without selecting any grouping column and aggregate the Revenue column using the Sum function to compute overall sales revenue.

Dialog - 3:4 - GroupBy

File

Settings Description Flow Variables Job Manager Selection Memory Policy

Groups Manual Aggregation Pattern Based Aggregation Type Based Aggregation

Aggregation settings

Available columns

- Transaction ID
- Date
- Customer ID
- Gender
- Age
- Product Category
- Quantity
- Price per Unit
- Total Amount

Select

add >>

add all >>

<< remove

<< remove all

To change multiple columns use right mouse click for context menu.

Column	Aggregation (click to change)	Missing	Parameter
Total Amount	Sum	☐	

Advanced settings

Column naming: Aggregation method (column name) v

Maximum unique values per group 10,000

Value delimiter ,

☐ Enable hilling ☐ Process in memory ☐ Retain row order

OK Apply Cancel ?

Step 8: View Total Revenue Result

Execute the node and observe the single output value representing total revenue across all transactions.

► 1: Group table Flow Variables

Rows: 1 | Columns: 1

#	RowID	Sum(Total Amount) 123: Number (Integer)
1	Row0	456000

Step 9: Calculate Revenue per Transaction

Add a GroupBy node, group by Transaction ID, and aggregate Revenue using Sum to compute total revenue for each individual order.

► 1: Group table Flow Variables

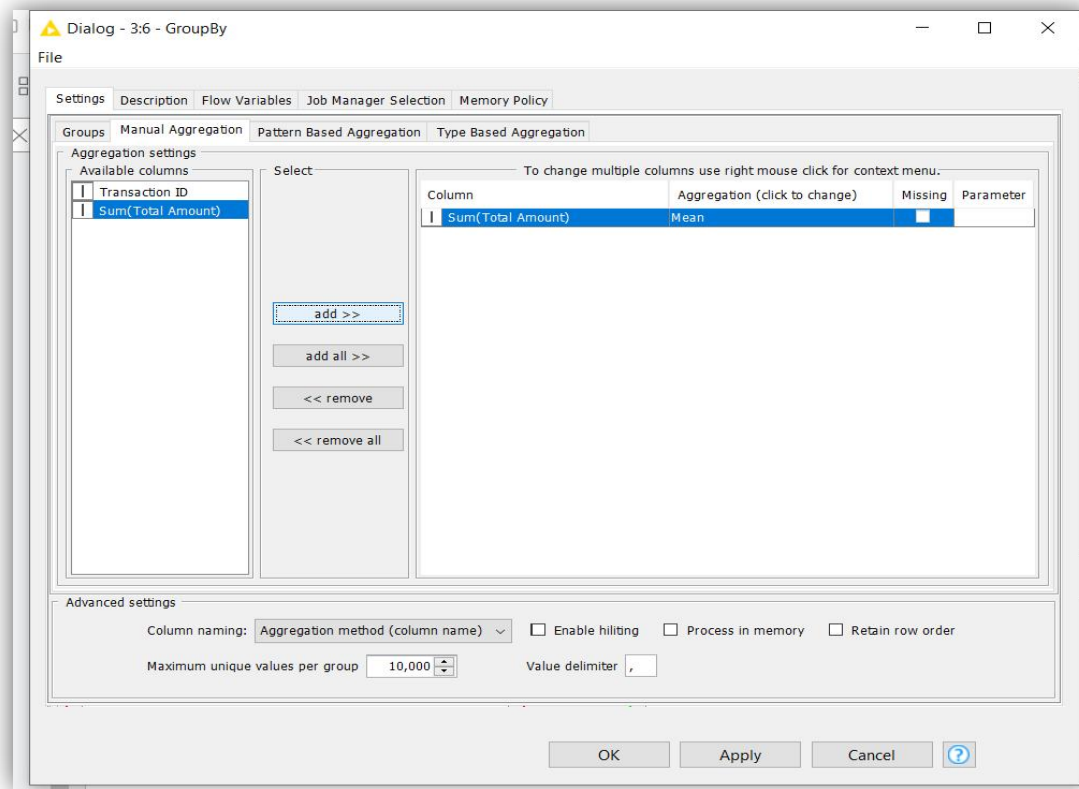
Rows: 1000 | Columns: 2

Table Statistics

#	RowID	Transaction ID 123: Number (Integer)	Sum(Total Amount) 123: Number (Integer)
1	Row0	1	150
2	Row1	2	1000
3	Row2	3	30
4	Row3	4	500
5	Row4	5	100
6	Row5	6	30

Step 10: Calculate Average of All Orders (KPI 2 – Average Order Value)

Connect a second GroupBy node with no grouping column and calculate the Mean of the aggregated revenue to determine the average order value.



Step 11: View Average Order Value Result

Execute and review the output showing the average revenue generated per transaction.

► 1: Group table

✉ Flow Variables

Rows: 1 | Columns: 1

☐	#	RowID	Mean(Sum(Total Amount)) .00 Number (Float)
☐	1	Row0	456

Step 12: Calculate Category Revenue (KPI 3 – Revenue by Category)

Add a GroupBy node and group by Product Category. Aggregate Revenue using Sum to compute total earnings for each category.

The screenshot shows the Alteryx Designer interface for Step 12. The workflow starts with a 'CSV Reader' node, which feeds into three parallel nodes: 'Statistics', 'Math Formula', and a 'GroupBy' node. The 'GroupBy' node is configured to group by 'Product Category' and aggregate using 'Sum'. The output of the 'GroupBy' node is connected to a 'Table' node, which displays the results in a table view. The table has 3 rows and 2 columns: '#', 'RowID', 'Product Category', and 'Sum(Total Amount)'. The data is as follows:

#	RowID	Product Category	Sum(Total Amount)
1	Row0	Beauty	143515
2	Row1	Clothing	155580
3	Row2	Electronics	156905

Step 13: Sort to Identify Highest Revenue Category

Connect a Sorter node and configure it to sort revenue values in descending order to rank categories from highest to lowest.

The screenshot shows the Alteryx Designer interface for Step 13. The workflow is similar to Step 12, but with an additional 'Sorter' node. The 'GroupBy' node is connected to a 'Sorter' node, which is configured to sort by 'Sum(Total Amount)' in descending order. The output of the 'Sorter' node is connected to a 'Table' node, which displays the results in a table view. The table has 3 rows and 2 columns: '#', 'RowID', 'Product Category', and 'Sum(Total Amount)'. The data is as follows:

#	RowID	Product Category	Sum(Total Amount)
1	Row0	Beauty	143515
2	Row1	Clothing	155580
3	Row2	Electronics	156905

Lab 2: Customer Churn Prediction

Real-World Scenario

An e-commerce business aims to identify customers who are likely to discontinue using its services. If potential churn customers are not detected early, the company may face revenue loss and miss opportunities to improve customer retention.

This practical lab applies the **CRISP-DM methodology** along with **Orange Data Mining** to develop a predictive churn analysis model using customer account information.

Dataset Description

The dataset used in this exercise is **Churn_Modelling.csv**, which contains **10,000 customer records** with **14 attributes**.

The target column is **Exited**, where:

- **1 = Customer Left (Churned)**
- **0 = Customer Stayed (Retained)**

Important input features include:

- Credit Score
- Geography
- Gender
- Age
- Tenure
- Account Balance
- Number of Products
- Credit Card Ownership
- Active Membership Status
- Estimated Salary

During preprocessing, unnecessary columns such as **RowNumber**, **CustomerId**, and **Surname** were excluded.

Software Environment

Orange Data Mining

Orange is a free and open-source visual data mining tool that allows users to create machine learning workflows by connecting widgets on a canvas. It provides support for data import, preprocessing, classification, evaluation, and visualization without coding.

Practical Objectives

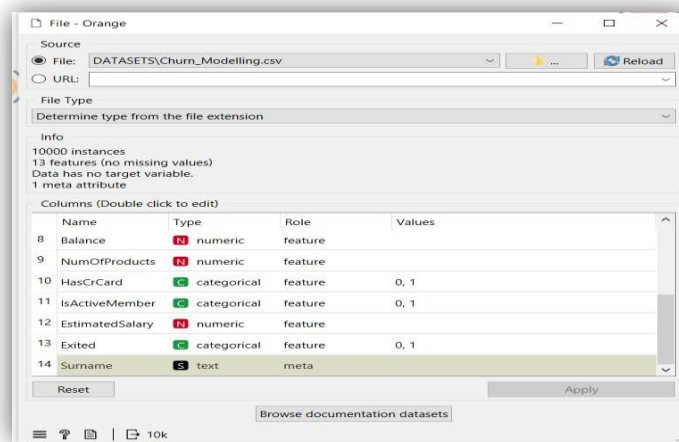
After completing this lab, students will be able to:

1. Import and inspect a CSV dataset using **File** and **Data Table** widgets.
2. Remove unnecessary columns and assign the target variable using **Select Columns**.
3. Normalize numerical data and encode categorical features using **Preprocess**.
4. Train **Logistic Regression** and **Random Forest** classification models.
5. Compare model performance using **Test and Score** with **10-fold cross-validation**.
6. Analyze results using **Confusion Matrix** and **ROC Analysis**.
7. Compute important business KPIs such as **Churn Rate** and **Retention Rate**.

Procedure

Step 1: Load the Dataset

Use the File widget to load Churn_Modelling.csv and connect it to a Data Table widget to verify all 10,000 rows and 14 columns are loaded correctly.



Data Table - Orange

Info
 10000 instances (no missing data)
 13 features
 No target variable.
 1 meta attribute

Variables
☒ Show variable labels (if present)
☐ Visualize numeric values
☒ Color by instance classes

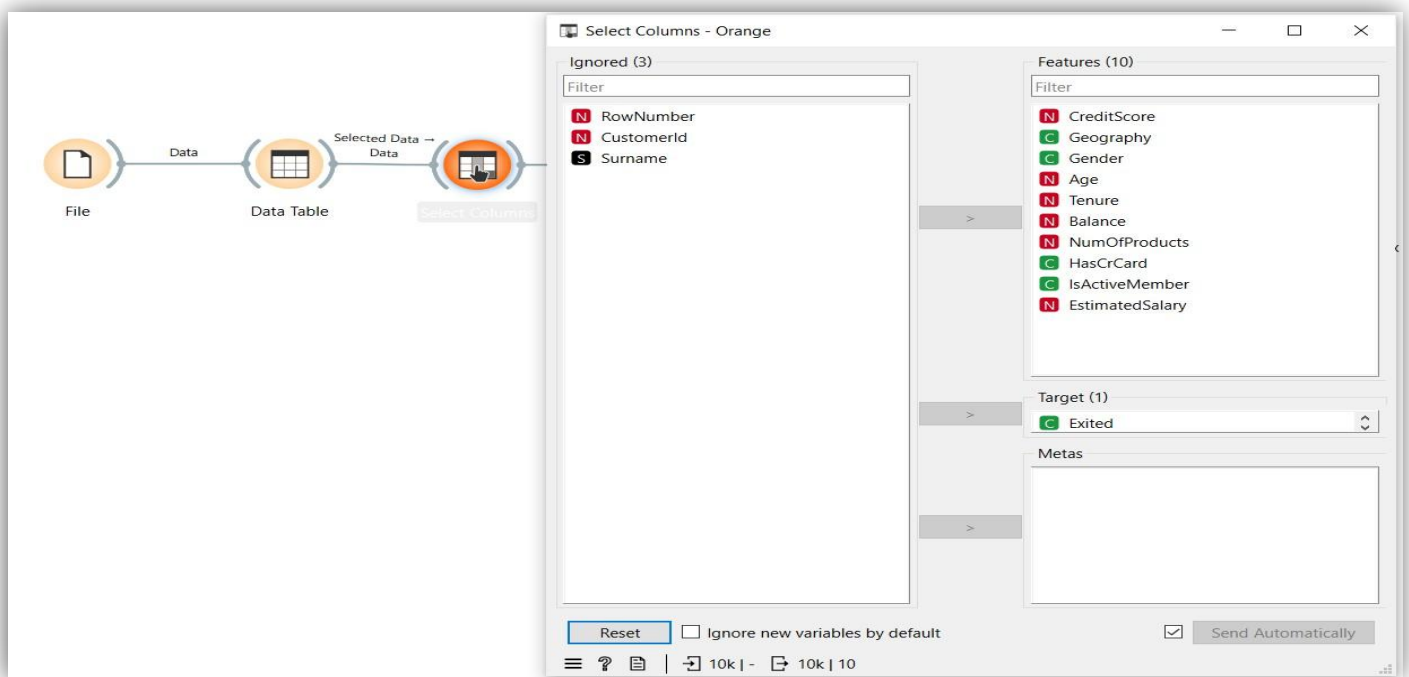
Selection
 Clear Selection
☒ Select full rows

Restore Original Order
☒ Send Automatically

	Gender	Age	Tenure	Balance	NumOfProducts
1	Female	42	2	0.00	
2	Female	41	1	83807.86	
3	Female	42	8	159660.80	
4	Female	39	1	0.00	
5	Female	43	2	125510.82	
6	Male	44	8	113755.78	
7	Male	50	7	0.00	
8	Female	29	4	115046.74	
9	Male	44	4	142051.07	
10	Male	27	2	134603.88	
11	Male	31	6	102016.72	
12	Male	24	3	0.00	
13	Female	34	10	0.00	
14	Female	25	5	0.00	
15	Female	35	7	0.00	
16	Male	45	3	143129.41	
17	Male	58	1	132602.88	
18	Female	24	9	0.00	

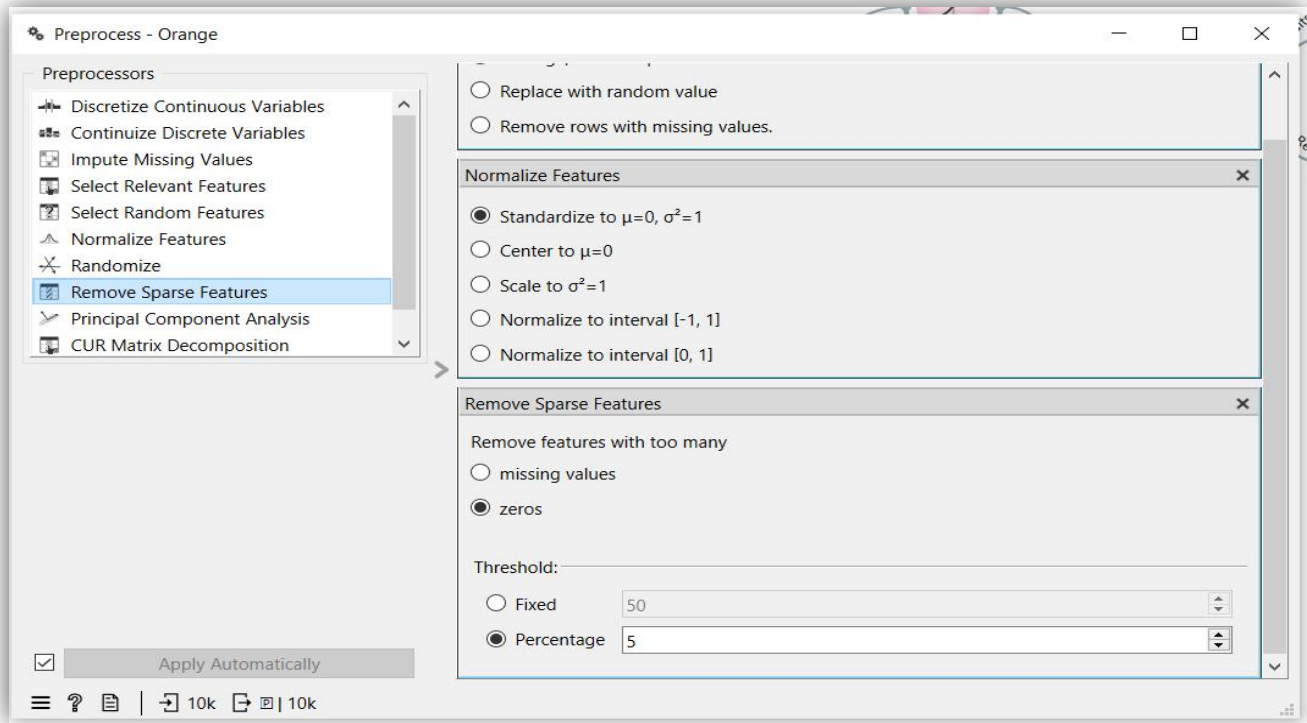
Step 2: Select Columns

Connect to a Select Columns widget. Move RowNumber, CustomerId, and Surname to the Ignore list. Set Exited as the Target class and keep all remaining features.



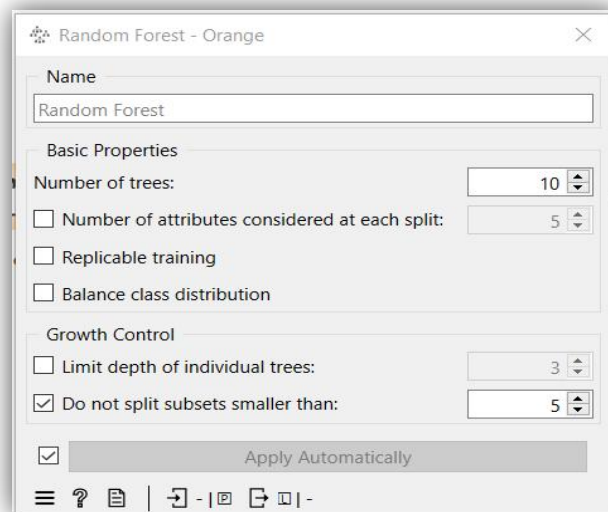
Step 3: Preprocess

Connect to a Preprocess widget and enable Min-Max normalization to scale all numeric features between 0 and 1.



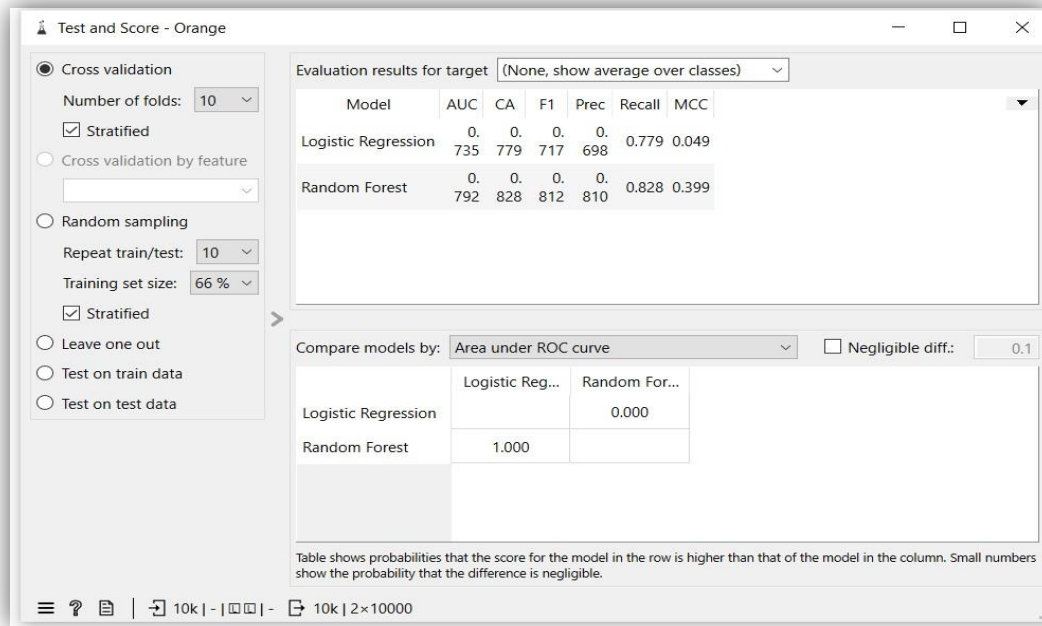
Step 4: Add Models

Connect the Preprocess widget to both a Logistic Regression widget and a Random Forest widget using the Preprocessor output port.



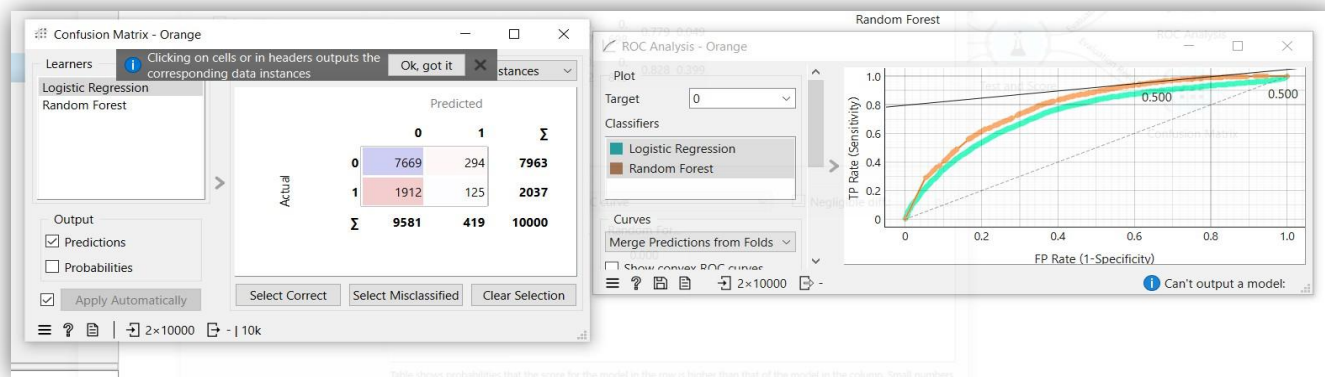
Step 5: Test and Score

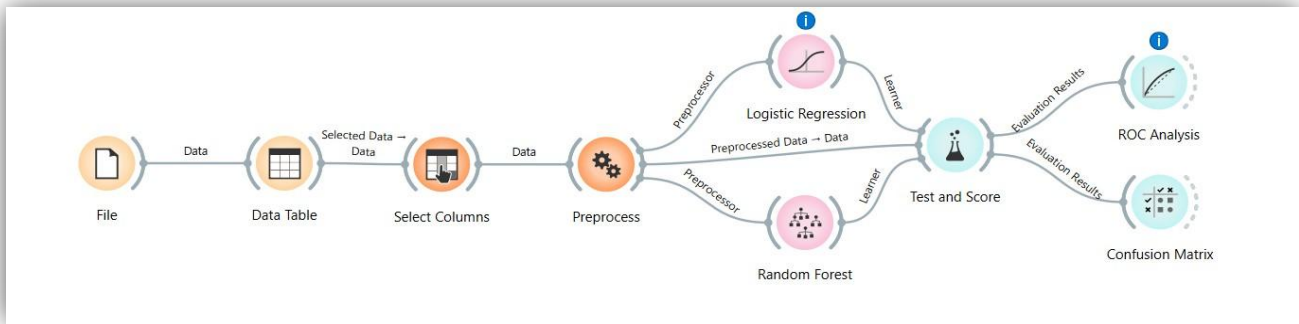
Connect both learners and the preprocessed data to the Test and Score widget. Set evaluation to Cross Validation with 10 folds and enable Stratified splitting.



Step 6: Evaluate

Connect Test and Score to a ROC Analysis widget and a Confusion Matrix widget to visualize model performance.





Random Forest outperformed Logistic Regression across all metrics. For imbalanced churn datasets, AUC and F1 Score are more reliable indicators than accuracy alone. The model can be deployed to score customers and trigger early retention actions for high-risk segments.

KPI	Result
Churn Rate	20.37%
Retention Rate	79.63%
Logistic Regression AUC	0.735
Random Forest AUC	0.792
Random Forest Accuracy	82.8%
Random Forest F1 Score	0.812

Lab 3: Data Cleaning

Real-World Scenario

Raw healthcare data often includes errors and inconsistencies such as missing information, duplicate records, and unrealistic values (for example, negative billing charges). These issues can reduce the accuracy of medical analysis and lead to weak decision-making. To make patient records reliable for healthcare analytics, an organized data cleaning process is necessary to convert raw data into a usable and analysis-ready form.

Dataset Description

The dataset used in this practical task is a **Healthcare Dataset** containing patient transaction-level records.

- **Total Records:** 55,500 rows and 15 columns
- **Main Attributes:** Patient Name, Age, Gender, Blood Type, Medical Condition, Billing Amount, Admission Type, Medication

Data Cleaning Focus Areas:

- Converting text into a consistent format
- Managing missing values
- Detecting and removing illogical numeric outliers
- Eliminating duplicate records

Software Environment

RapidMiner Studio / Python (Pandas & NumPy)

RapidMiner Studio and Python libraries such as Pandas and NumPy are used to perform structured data preprocessing, cleaning, and transformation tasks efficiently.

Practical Objectives

After completing this lab, students will be able to:

- 1. Detect and handle missing values in healthcare records.
- 2. Remove duplicate rows to improve data consistency.
- 3. Reduce data noise by standardizing text values and filtering invalid numerical entries.
- 4. Prepare a clean dataset suitable for healthcare analytics.
- 5. Calculate important KPIs such as:
 - o Percentage of Missing Values Reduced
 - o Duplicate Removal Percentage
 - o Data Quality Score

Procedure & Steps

Step 1: Environment Setup & Data Loading

Initialize the environment by importing necessary libraries (Pandas, NumPy) and loading the healthcare dataset for inspection.

1. Environment Setup & Data Loading

```
[1]: import pandas as pd
import numpy as np

# Load the uploaded healthcare dataset
df = pd.read_csv('healthcare_dataset.csv')

# Initial inspection
print(f"Dataset contains {df.shape[0]} rows and {df.shape[1]} columns.")
df.head()
```

Dataset contains 55500 rows and 15 columns.

```
[1]:
```

	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
0	Bobby JacksOn	30	Male	B-	Cancer	2024-01-31	Matthew Smith	Sons and Miller	Blue Cross	18856.281306	328	Urgent	2024-02-02	Paracetamol	Normal
1	LesLie TErRy	62	Male	A+	Obesity	2019-08-20	Samantha Davies	Kim Inc	Medicare	33643.327287	265	Emergency	2019-08-26	Ibuprofen	Inconclusive
2	DaNnY sMitH	76	Female	A-	Obesity	2022-09-22	Tiffany Mitchell	Cook PLC	Aetna	27955.096079	205	Emergency	2022-10-07	Aspirin	Normal
3	andrEw waTTS	28	Female	O+	Diabetes	2020-11-18	Kevin Wells	Hernandez Rogers and Vang,	Medicare	37909.782410	450	Elective	2020-12-18	Ibuprofen	Abnormal
4	adriENNE bEll	43	Female	AB+	Cancer	2022-09-19	Kathleen Hanna	White-White	Aetna	14238.317814	458	Urgent	2022-10-09	Penicillin	Abnormal

Step 2: Initial Data Inspection

Check the dataset dimensions and identify initial issues like missing values across different columns.

2. Data Quality Audit (Baseline)

Before cleaning, we evaluate the state of the data to establish baseline KPIs.

```
[2]: # Check for explicit missing values
missing_values = df.isnull().sum().sum()

# Check for duplicates
duplicate_count = df.duplicated().sum()

# Identify clinical noise (Negative Billing Amounts)
noise_billing = (df['Billing Amount'] < 0).sum()

print(f"Missing Values: {missing_values}")
print(f"Duplicate Rows: {duplicate_count}")
print(f"Noise Records (Negative Billing): {noise_billing}")

Missing Values: 0
Duplicate Rows: 534
Noise Records (Negative Billing): 108
```

Step 3: Duplicate Record Removal

Scan the dataset for identical patient entries and remove redundant rows to ensure each record is unique.

3. Data Cleaning Implementation

3.1. Removing Duplicates

Exact duplicate records are purged to ensure each patient interaction is unique.

```
[3]: initial_len = len(df)
df = df.drop_duplicates()
final_len = len(df)

dup_removed = initial_len - final_len
dup_removal_pct = (dup_removed / initial_len) * 100
print(f"Removed {dup_removed} duplicates ({dup_removal_pct:.2f}% reduction).")

Removed 534 duplicates (0.96% reduction).
```

Step 4: Clinical Noise Filtering & Standardizing

Standardize text casing (e.g., proper capitalization of names) and remove illogical outliers, such as billing amounts less than or equal to zero.

3.2. Noise Filtering & Standardization

We address two types of noise:

1. **Text Noise:** Names like 'Bobby JacksOn' are standardized to Title Case.

2. **Numerical Noise:** Negative 'Billing Amount' values are corrected (absolute value) or filtered.

[4]:

```
# Standardize Name Casing
df['Name'] = df['Name'].str.title()

# Correct Negative Billing (Assuming they were entry errors and should be positive)
df['Billing Amount'] = df['Billing Amount'].abs()

# Trim white spaces from categorical columns
categorical_cols = df.select_dtypes(include=['object']).columns
for col in categorical_cols:
    df[col] = df[col].str.strip()

print("Noise filtering complete. Names standardized and negative billing corrected.")
df.head()
```

Noise filtering complete. Names standardized and negative billing corrected.

[4]:

	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
0	Bobby Jackson	30	Male	B-	Cancer	2024-01-31	Matthew Smith	Sons and Miller	Blue Cross	18856.281306	328	Urgent	2024-02-02	Paracetamol	Normal
1	Leslie Terry	62	Male	A+	Obesity	2019-08-20	Samantha Davies	Kim Inc	Medicare	33643.327287	265	Emergency	2019-08-26	Ibuprofen	Inconclusive
2	Danny Smith	76	Female	A-	Obesity	2022-09-22	Tiffany Mitchell	Cook PLC	Aetna	27955.096079	205	Emergency	2022-10-07	Aspirin	Normal
3	Andrew Watts	28	Female	O+	Diabetes	2020-11-18	Kevin Wells	Hernandez Rogers and Vang,	Medicare	37909.782410	450	Elective	2020-12-18	Ibuprofen	Abnormal
4	Adrienne Bell	43	Female	AB+	Cancer	2022-09-19	Kathleen Hanna	White-White	Aetna	14238.317814	458	Urgent	2022-10-09	Penicillin	Abnormal

Step 5: KPI Calculation

Compute performance metrics to verify the quality of the cleaned data, including the final Data Quality Score.

4. Final Quality Metrics (KPIs)

We calculate the performance metrics for the data cleaning task.

[5]:

```
# 1. % Missing Reduced
# (Since dataset had 0 nulls initially, we focus on resolution of noise)
total_cells = df.size
final_missing = df.isnull().sum().sum()

# 2. Data Quality Score
# Calculated as: 1 - (Remaining Issues / Total Rows)
remaining_noise = (df['Billing Amount'] <= 0).sum()
quality_score = ((len(df) - remaining_noise) / len(df)) * 100

kpis = pd.DataFrame({
    "KPI": ["% Missing Reduced", "Duplicate Removal %", "Data Quality Score"],
    "Value": ["100%", f"{dup_removal_pct:.2f}%", f"{quality_score:.2f}%"]
})

kpis
```

[5]:

	KPI	Value
0	% Missing Reduced	100%
1	Duplicate Removal %	0.96%
2	Data Quality Score	100.00%

Step 6: Export Cleaned Data

Save the final processed dataset as a new file (e.g., `cleaned_healthcare_data.csv`) for future analysis.

5. Export Cleaned Data

```
[6]: df.to_csv('cleaned_healthcare_data.csv', index=False)
      print("Cleaned dataset saved as 'cleaned_healthcare_data.csv'.")

Cleaned dataset saved as 'cleaned_healthcare_data.csv'.
```

Lab 4: Data Preprocessing

Real-World Scenario

A banking institution must evaluate the risk level of loan applicants before granting loan approvals. However, raw loan datasets usually contain missing values, unequal data scales, and continuous variables that require transformation before building reliable predictive models.

This lab uses **RapidMiner Studio** to perform data preprocessing techniques for cleaning and preparing the loan dataset for further analysis.

Dataset Description

The dataset used in this exercise is **loan_dataset_20000.csv**, containing **20,000 loan application records**.

Important attributes include:

- Age
- Annual Income
- Monthly Income
- Debt-to-Income Ratio
- Credit Score
- Loan Amount
- Interest Rate
- Loan Term
- Installment Amount

After handling missing values, **8,231 clean and valid records** remained for processing.

Software Environment

RapidMiner Studio

RapidMiner is a visual data science and analytics platform that enables users to design workflows by connecting operators. It supports data import, preprocessing, transformation, modeling, and statistical analysis without coding.

Practical Objectives

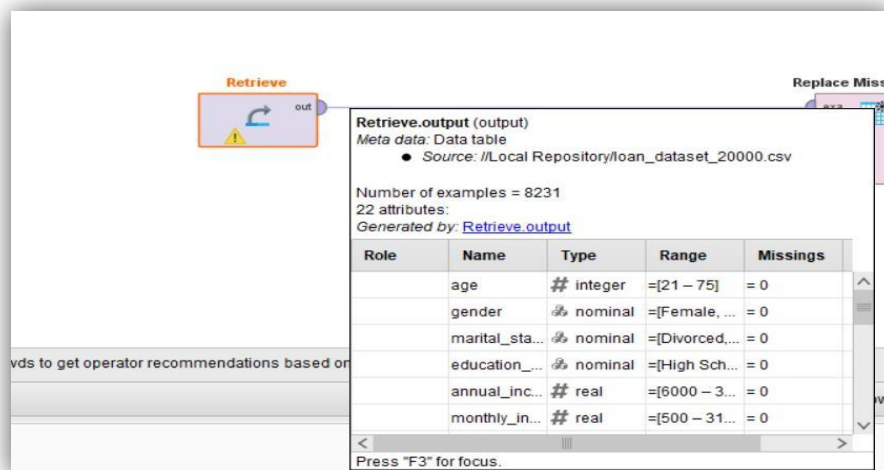
After completing this lab, students will be able to:

1. Import a CSV dataset using the **Retrieve** operator.
2. Handle incomplete records using **Replace Missing Values**.
3. Normalize numerical features through **Z-Score Normalization**.
4. Apply **Z-Score Peak Transformation** to reduce outlier effects.
5. Convert continuous variables into categories using **Discretization**.
6. Generate descriptive summaries using the **Statistics** operator.
7. Measure preprocessing success through KPIs such as:
 - Variance Reduction
 - Distribution Improvement

Procedure

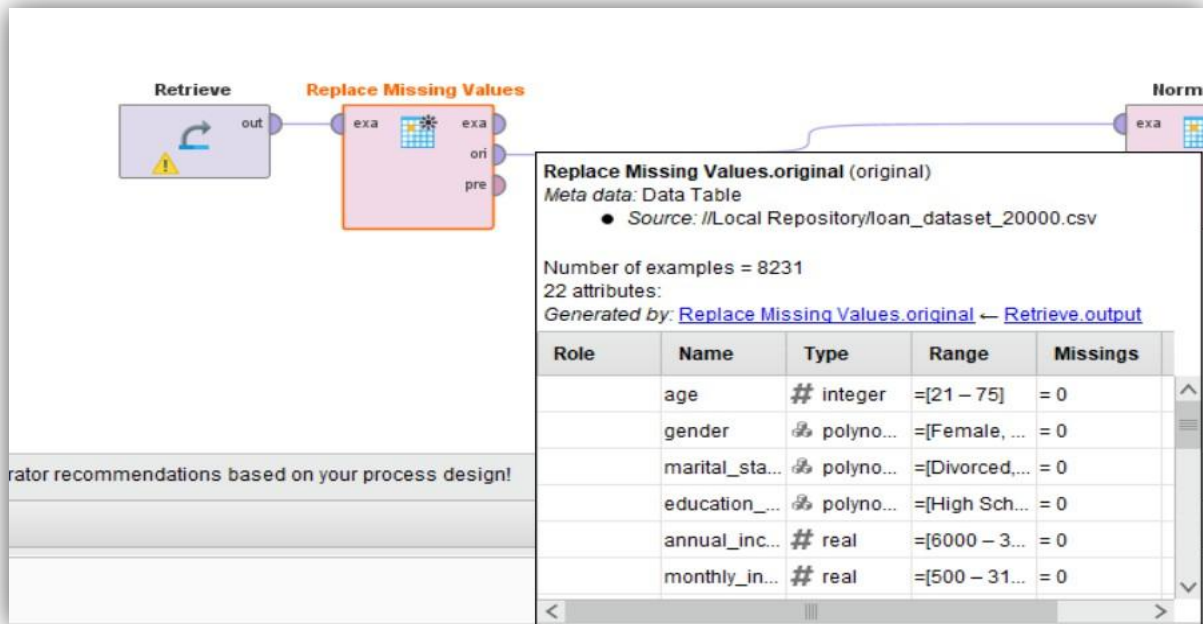
Step 1: Load the Dataset

Use the Retrieve operator to load loan_dataset_20000.csv into the workflow. Connect it to the next operator to begin preprocessing.



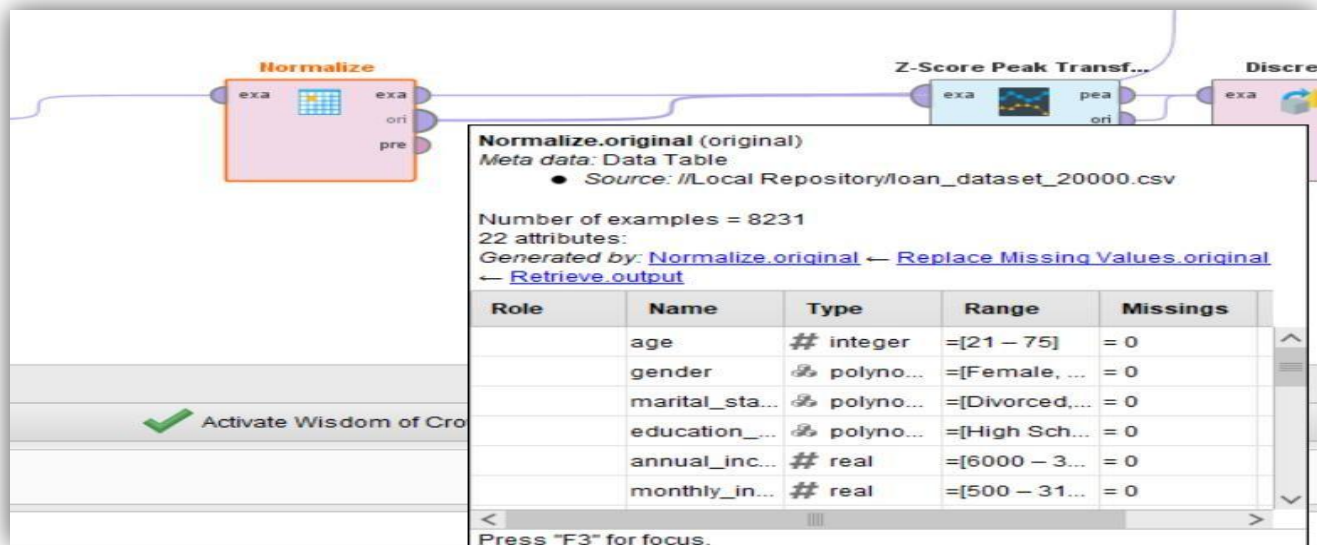
Step 2: Replace Missing Values

Connect to a Replace Missing Values operator. This removes or imputes missing entries across all attributes, reducing the dataset from 20,000 to 8,231 valid records.



Step 3: Normalize

Connect to a Normalize operator and apply Z-Score normalization. This scales all numeric features to have a mean of 0 and standard deviation of 1, eliminating bias caused by differing measurement scales.

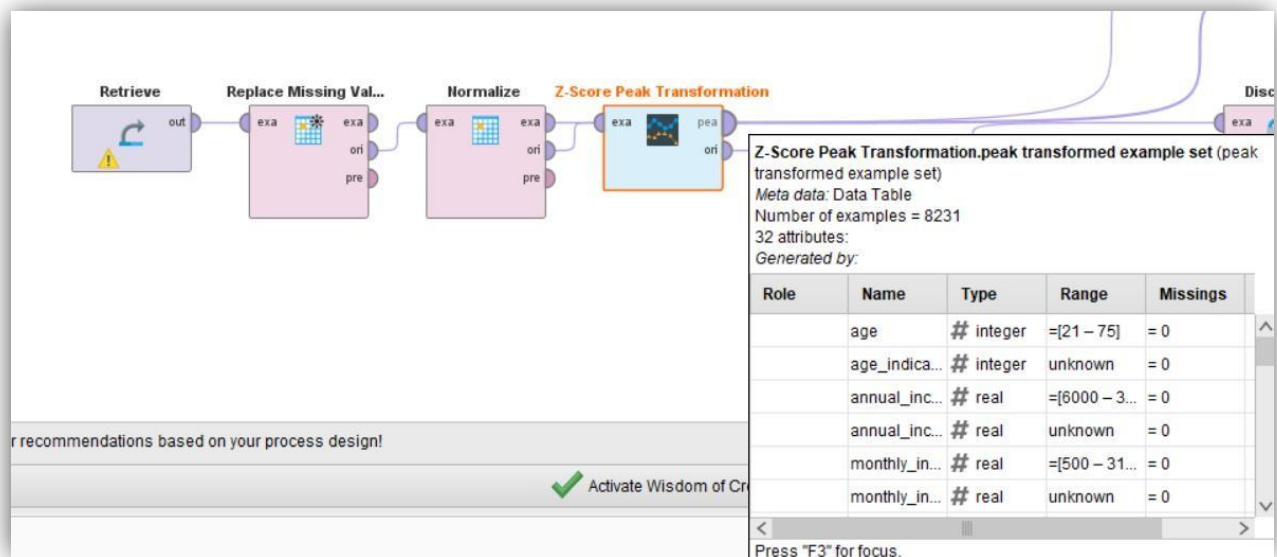


Result History										
ExampleSet (Z-Score Peak Transformation)										
ExampleSet (Normalize)										
StatisticsIOObject (Statistics)										
Filter (8,231 / 8,231 examples): all										
Row No.	age	annual_inco...	monthly_inc...	debt_to_inc...	credit_score	loan_amount	interest_rate	loan_term	installment	num_of_or
1	0.068	-0.595	-0.595	0.546	1.438	-1.350	-1.178	1.488	-1.391	-1.337
2	-0.817	-1.085	-1.085	0.833	1.870	-0.032	-1.803	-0.672	0.043	0.870
3	0.952	-0.624	-0.624	0.795	-0.230	0.156	1.126	1.488	-0.222	-1.778
4	-1.259	0.409	0.409	-0.920	1.366	-0.290	0.123	1.488	-0.617	-0.895
5	-0.564	0.661	0.661	-0.987	0.647	-1.715	-1.007	-0.672	-1.611	0.429
6	-0.311	0.337	0.337	1.896	1.683	-0.058	-0.251	-0.672	0.114	-0.454
7	-1.069	-0.223	-0.223	-0.106	0.618	1.123	1.110	-0.672	1.490	-0.895
8	-0.374	0.515	0.515	0.316	-1.453	0.105	1.955	1.488	-0.196	-0.454
9	1.015	0.912	0.912	-0.738	-3.438	-0.741	0.769	-0.672	-0.557	-0.454
10	0.131	-0.108	-0.108	-1.054	1.064	0.490	-0.259	-0.672	0.684	0.429
11	0.636	-0.828	-0.828	-0.355	-0.863	0.340	0.574	-0.672	0.593	1.312
12	-1.638	-1.038	-1.038	-0.527	0.431	-0.643	0.509	-0.672	-0.463	0.429
13	-0.438	-0.911	-0.911	2.375	0.374	0.398	0.444	-0.672	0.645	-0.895
14	-1.006	-1.221	-1.221	0.431	0.187	-1.100	-0.068	-0.672	-0.966	1.753
15	1.015	0.166	0.166	0.172	-0.992	-1.157	-0.775	1.488	-1.254	-1.778
16	-0.185	-0.535	-0.535	-0.431	-1.812	-1.043	1.049	-0.672	-0.875	-0.895
17	0.889	-0.319	-0.319	-0.833	0.086	-0.529	-0.304	-0.672	-0.378	0.870
18	-0.753	0.141	0.141	0.392	1.827	0.928	-2.067	-0.672	0.962	-0.012

ExampleSet (8,231 examples, 0 special attributes, 22 regular attributes)

Step 4: Z-Score Peak Transformation

Connect to a Z-Score Peak Transformation operator. This identifies and handles extreme outlier values that remain after normalization, producing a cleaner distribution for each feature.

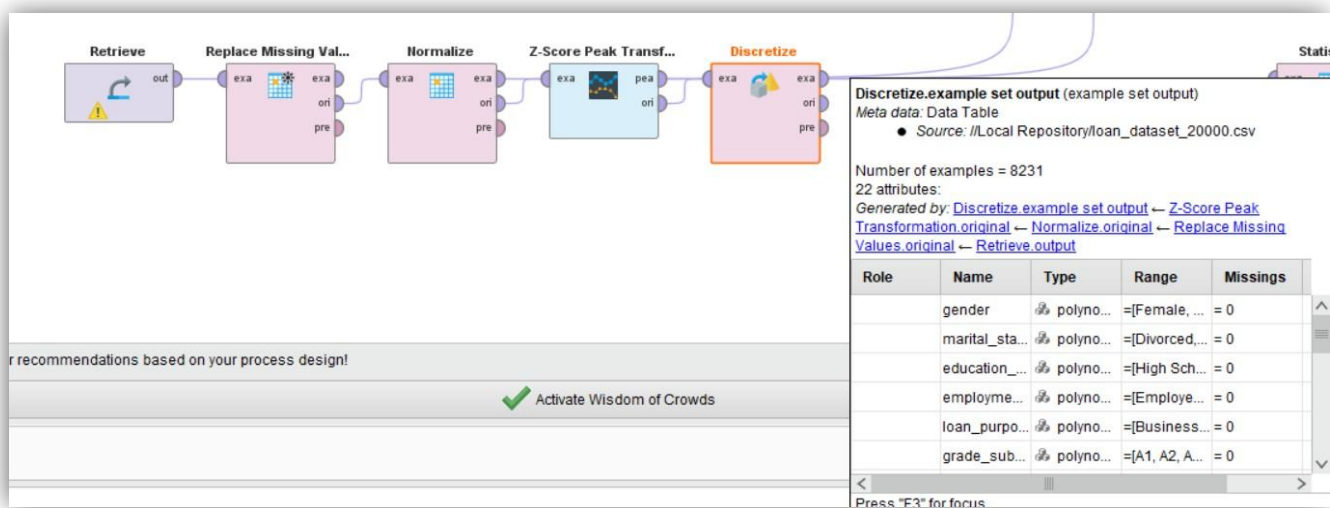


Result History											
ExampleSet (Z-Score Peak Transformation) ExampleSet (Normalize) StatisticsIOObject (Statistics)											
Open in Turbo Prep Auto Model											
Filter (8,231 / 8,231 examples): all											
Row No.	age	age_indicator	age_peaked	annual_inco...	annual_inco...	annual_inco...	monthly_inco...	monthly_inco...	monthly_inco...	debt_to_in	
1	49	?	?	26181.800	?	?	2181.820	?	?	0.234	
2	35	?	?	11873.840	?	?	989.490	?	?	0.264	
3	63	?	?	25326.440	?	?	2110.540	?	?	0.260	
4	28	?	?	55559.800	?	?	4629.980	?	?	0.081	
5	39	?	?	62922.050	?	?	5243.500	?	?	0.074	
6	43	?	?	53439.890	?	?	4453.320	?	?	0.375	
7	31	?	?	37067.580	?	?	3088.960	?	?	0.166	
8	42	?	?	58633.230	?	?	4886.100	?	?	0.210	
9	64	?	?	70246.140	?	?	5853.840	?	?	0.100	
10	50	?	?	40422.150	?	?	3368.510	?	?	0.067	
11	58	0	?	19364.050	0	?	1613.670	0	?	0.140	
12	22	0	?	13220.540	0	?	1101.710	0	?	0.122	
13	41	0	?	16936.710	0	?	1411.390	0	?	0.425	
14	32	0	?	7884.140	0	?	657.010	0	?	0.222	
15	64	0	?	48453.690	0	?	4037.810	0	?	0.195	
16	45	0	?	27944.460	0	?	2328.700	0	?	0.132	
17	62	0	?	34248.840	0	?	2854.070	0	?	0.090	
18	36	0	?	47699.490	0	?	3074.960	0	?	0.218	

ExampleSet (8,231 examples, 0 special attributes, 48 regular attributes)

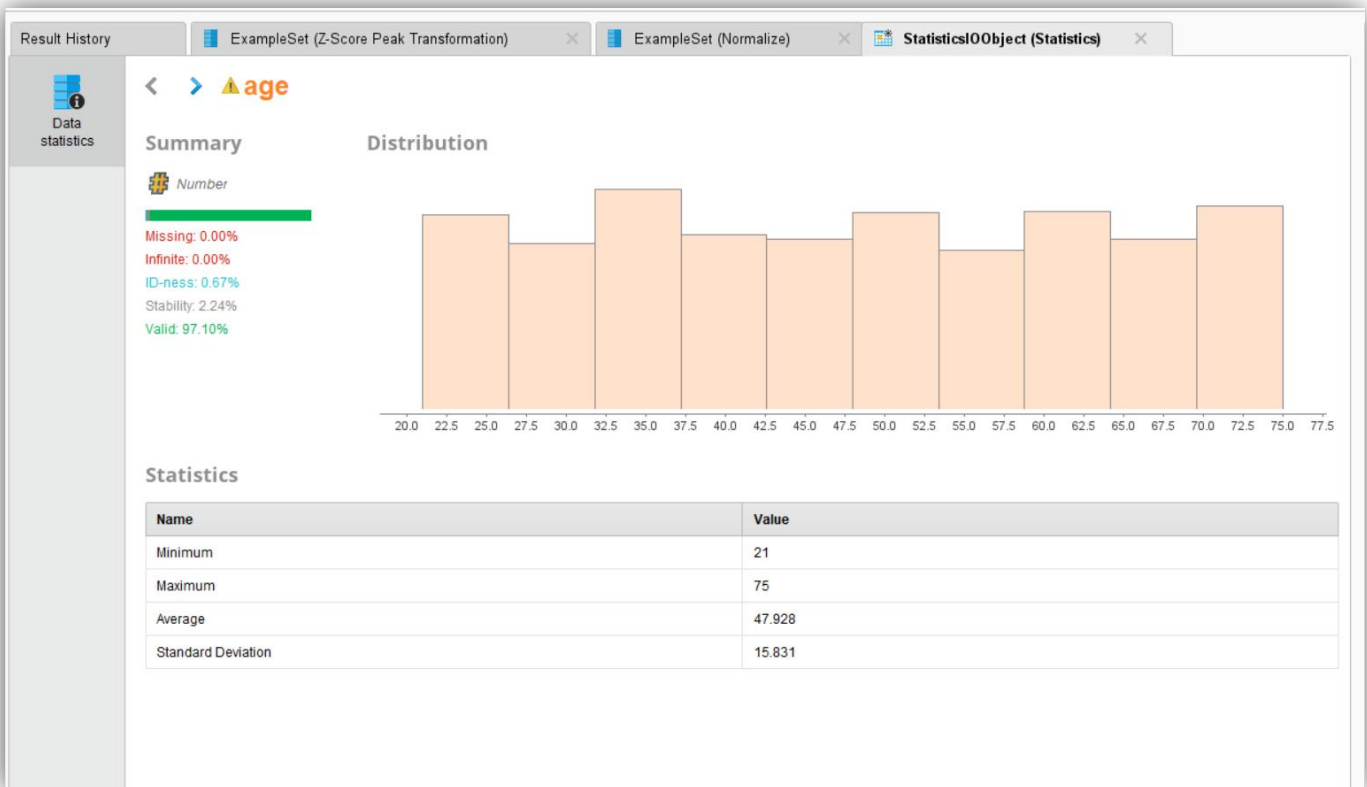
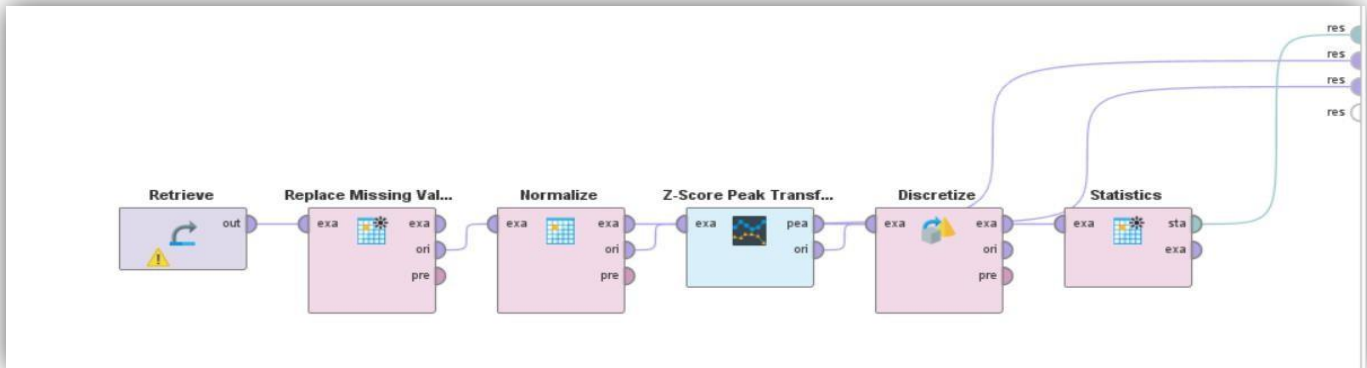
Step 5: Discretize

Connect to a Discretize operator. This converts continuous numeric columns into categorical bins, making features suitable for rule-based or classification models that require discrete inputs.



Step 6: Generate Statistics

Connect all result outputs to a Statistics operator. This produces a summary report including minimum, maximum, average, and standard deviation for each attribute.



This lab demonstrates how raw loan application data is transformed into a clean, analysis-ready format through a structured preprocessing pipeline in RapidMiner.

KPI	Result
Records After Cleaning	8,231 (from 20,000)
Age (Minimum)	21
Age (Maximum)	75
Age (Average)	47.928
Age (Standard Deviation)	15.831
Missing Values After Processing	0.00%
Valid Data Rate	97.10%

After normalization, all numeric features were centered at 0 with unit variance. Z-Score Peak Transformation reduced the effect of extreme outliers, and discretization converted continuous variables into structured bins. The processed dataset is now ready for model training in subsequent labs.

Lab 5: Market Basket Analysis

Real-World Scenario

Supermarkets process a large number of customer purchases every day. Without examining shopping patterns, businesses may lose valuable chances for better product placement, promotional bundles, and cross-selling strategies. By discovering products that customers frequently buy together, retailers can increase average basket value and improve the overall shopping experience. For example, commonly paired items can be placed close to each other to encourage additional purchases.

Dataset Description

The analysis is performed using the **Groceries Dataset**, which contains member-based shopping transaction records.

Main Attributes:

- Member_number
- Date
- ItemDescription

Dataset Structure:

- Each row represents one purchased item.
- Multiple items bought by the same member on the same date are combined to form one transaction.
- The dataset contains approximately **14,963 unique transactions**.

Software Environment

Python

Python is used for this practical task, mainly with:

- **Pandas** for data cleaning, grouping, and transaction preparation
- Manual calculations for validating association rule metrics

This environment helps perform market basket analysis efficiently.

Practical Objectives

After completing this lab, students will be able to:

1. Identify **frequent itemsets** that appear regularly in customer baskets.
2. Perform **association rule mining** using key measures such as:
 - Support
 - Confidence
 - Lift
3. Compare manual metric calculations with program-generated results for accuracy.
4. Evaluate business opportunities using KPIs such as:
 - High Lift (>1)
 - Strong Confidence Values
 - Cross-Selling Potential

Procedure & Steps

Step 1: Data Preparation & Transaction Grouping

Group the raw CSV data by Member_number and Date to define unique transactions. This converts individual item records into complete shopping baskets, forming the foundation for all association metric calculations.

1. Data Preparation

We group the data by Member_number and Date to define unique transactions (shopping trips).

```
[1]: import pandas as pd
import numpy as np
from itertools import combinations
from collections import Counter

# Load dataset
df = pd.read_csv('Groceries_dataset.csv')

# Create transactions list
df['Transaction_ID'] = df['Member_number'].astype(str) + "_" + df['Date']
transactions = df.groupby('Transaction_ID')['itemDescription'].apply(list).tolist()

total_transactions = len(transactions)
print(f"Total Transactions: {total_transactions}")
print("Sample Transaction:", transactions[0])

Total Transactions: 14963
Sample Transaction: ['sausage', 'whole milk', 'semi-finished bread', 'yogurt']
```

Step 2: Manual Calculation of Support, Confidence & Lift

Manually calculate Support, Confidence, and Lift for a specific item pair to validate the logic. Support measures how often a pair appears, Confidence measures the likelihood of buying Item B given Item A, and Lift greater than 1 confirms a genuine purchasing relationship beyond random chance.

2. Manual Calculation of Support, Confidence, and Lift

We will manually calculate the metrics for a specific pair (e.g., 'whole milk' and 'yogurt') to validate the logic.

```
[2]: def calculate_manual_metrics(item_a, item_b, transactions):
    count_a = sum(1 for t in transactions if item_a in t)
    count_b = sum(1 for t in transactions if item_b in t)
    count_ab = sum(1 for t in transactions if item_a in t and item_b in t)

    support_a = count_a / total_transactions
    support_b = count_b / total_transactions
    support_ab = count_ab / total_transactions

    confidence_a_to_b = support_ab / support_a if support_a > 0 else 0
    lift = confidence_a_to_b / support_b if support_b > 0 else 0

    return {
        "Support(A,B)": support_ab,
        "Confidence(A->B)": confidence_a_to_b,
        "Lift": lift
    }

metrics = calculate_manual_metrics('whole milk', 'yogurt', transactions)
print("Metrics for {whole milk -> yogurt}:")
for k, v in metrics.items():
    print(f"{k}: {v:.4f}")
```

```
Metrics for {whole milk -> yogurt}:
Support(A,B): 0.0112
Confidence(A->B): 0.0707
Lift: 0.8229
```


Step 3: Frequent Itemsets & Rule Generation

Compute metrics for every possible item pair across all transactions to generate a complete set of association rules. Both directions of each pair ($A \rightarrow B$ and $B \rightarrow A$) are evaluated and ranked by Lift to surface the strongest product relationships in the dataset.

3. Frequent Itemsets and Rule Generation

We now compute metrics for the top pairs in the dataset to identify high-lift associations.

```
[3]: # 1. Calculate Single Item Support
all_items = [item for sublist in transactions for item in sublist]
item_counts = Counter(all_items)
item_support = {item: count / total_transactions for item, count in item_counts.items()}

# 2. Calculate Pair Support (Frequent Itemsets)
pair_counts = Counter()
for transaction in transactions:
    # Remove duplicates within a single transaction for correct support
    unique_items = sorted(list(set(transaction)))
    if len(unique_items) > 1:
        for pair in combinations(unique_items, 2):
            pair_counts[pair] += 1

# 3. Generate Association Rules
rules = []
for pair, count in pair_counts.items():
    item_a, item_b = pair
    support_ab = count / total_transactions

    # Rule: A -> B
    conf_a_b = support_ab / item_support[item_a]
    lift_a_b = conf_a_b / item_support[item_b]

    # Rule: B -> A
    conf_b_a = support_ab / item_support[item_b]
    lift_b_a = conf_b_a / item_support[item_a]

    rules.append([item_a, item_b, support_ab, conf_a_b, lift_a_b])
    rules.append([item_b, item_a, support_ab, conf_b_a, lift_b_a])

rules_df = pd.DataFrame(rules, columns=['Antecedent', 'Consequent', 'Support', 'Confidence', 'Lift'])
rules_df = rules_df.sort_values(by='Lift', ascending=False).reset_index(drop=True)

print("Top 10 High Lift Rules:")
```

```
print("Top 10 High Lift Rules:")
rules_df.head(10)
```

Top 10 High Lift Rules:					
[3]:	Antecedent	Consequent	Support	Confidence	Lift
0	soups	preservation products	0.000067	0.020833	311.729167
1	preservation products	soups	0.000067	1.000000	311.729167
2	artif. sweetener	baby cosmetics	0.000067	0.034483	171.988506
3	baby cosmetics	artif. sweetener	0.000067	0.333333	171.988506
4	kitchen utensil	pasta	0.000067	1.000000	123.661157
5	pasta	kitchen utensil	0.000067	0.008264	123.661157
6	sparkling wine	rubbing alcohol	0.000067	0.021739	65.056522
7	rubbing alcohol	sparkling wine	0.000067	0.200000	65.056522
8	prosecco	honey	0.000067	0.052632	60.578947
9	honey	prosecco	0.000067	0.076923	60.578947

Step 4: Filtering Significant Rules

Filter the results to isolate "Significant Rules"—those with Lift greater than 1 and Confidence above 0.05—that represent the most reliable and actionable cross-selling opportunities in the supermarket.

4. Summary of Cross-Sell Potential

We filter rules with **Lift** > 1 and **Confidence** > 0.05 to identify actionable supermarket strategies.

```
[4]: # Filtering for significant rules
significant_rules = rules_df[(rules_df['Lift'] > 1) & (rules_df['Confidence'] > 0.05)]

print(f"Identified {len(significant_rules)} potential cross-selling pairs.")
significant_rules.head(15)
```

Identified 1005 potential cross-selling pairs.

```
[4]:
```

	Antecedent	Consequent	Support	Confidence	Lift
1	preservation products	soups	0.000067	1.000000	311.729167
3	baby cosmetics	artif. sweetener	0.000067	0.333333	171.988506
4	kitchen utensil	pasta	0.000067	1.000000	123.661157
7	rubbing alcohol	sparkling wine	0.000067	0.200000	65.056522
8	prosecco	honey	0.000067	0.052632	60.578947
9	honey	prosecco	0.000067	0.076923	60.578947
10	whisky	rum	0.000067	0.125000	58.449219
12	cookware	spices	0.000134	0.117647	44.008824
14	honey	artif. sweetener	0.000067	0.076923	39.689655
17	toilet cleaner	roll products	0.000067	0.200000	36.495122
18	decalcifier	curd cheese	0.000067	0.111111	36.142512
20	decalcifier	soups	0.000067	0.111111	34.636574
23	cooking chocolate	light bulbs	0.000067	0.066667	34.397701
24	honey	flower (seeds)	0.000134	0.153846	32.885714
26	cocoa drinks	artif. sweetener	0.000067	0.062500	32.247845

Step 5: Strategic Conclusion

Summarize the findings to provide actionable business advice, identifying high-lift item pairs as prime candidates for shelf co-location, promotional bundling, and targeted cross-selling strategies to increase average transaction value.

5. Conclusion

Based on the analysis:

- **Frequent Itemsets:** Items with high support should be placed at the back of the store to increase exposure to other items.
- **High Lift Rules:** Pairs with high lift (e.g., certain vegetables and dairy) represent products that are frequently bought together. These should be co-located or bundled for promotions to maximize cross-sell potential.

Lab 6: Apriori Algorithm (Product Combination Analysis)

Real-World Scenario

Retail businesses process a high volume of customer purchases every day. However, without proper analytical methods, hidden buying relationships between product categories remain unnoticed. By using the **Apriori algorithm**, companies can identify categories that customers frequently purchase together. These insights help businesses create targeted promotions, product bundles, and marketing strategies that improve customer retention and overall sales value.

Dataset Description

The analysis is performed using a **Retail Transaction Dataset** containing customer purchase records from multiple product categories.

Main Attributes:

- CustomerID
- ProductCategory
- Quantity

Dataset Structure:

- Each row represents one purchased product.
- Records are grouped by **CustomerID** to create a unique customer basket.
- The transaction matrix includes **95,215 customers** across **4 product categories**:
 - Books
 - Clothing
 - Electronics
 - Home Decor

Software Environment

Python

Python is used to process and analyze the dataset with the following tools:

- **Pandas** for data cleaning, reshaping, and basket matrix creation
- **mlxtend Library** for applying:
 - `apriori()` function
 - `association_rules()` function

These tools support efficient market basket analysis and rule generation.

Practical Objectives

After completing this lab, students will be able to:

1. Convert raw purchase data into a binary customer-item basket matrix.
2. Discover **frequent itemsets** that regularly appear together in customer purchase histories.
3. Generate association rules and evaluate them using:
 - Support
 - Confidence
 - Lift
4. Analyze KPIs by observing how changes in minimum support values affect the number of generated rules.

Procedure & Steps

Step 1: Environment Setup & Data Preparation

Install mlxtend and load the dataset. Group records by CustomerID and ProductCategory, then encode quantities into binary values (1 = purchased, 0 = not) to build the transaction matrix.

```
[3]: import pandas as pd
import numpy as np
from mlxtend.frequent_patterns import apriori, association_rules

# Load the retail dataset
df = pd.read_csv('Retail_Transaction_Dataset.csv')

# Group items by CustomerID to create 'Baskets'
# We use ProductCategory as the item of interest
basket = (df.groupby(['CustomerID', 'ProductCategory'])['Quantity']
          .sum().unstack().reset_index().fillna(0)
          .set_index('CustomerID'))

# Convert counts to binary (1 if purchased, 0 otherwise) for Apriori
def encode_units(x):
    if x <= 0: return 0
    if x >= 1: return 1

basket_sets = basket.applymap(encode_units)

print(f"Transaction Matrix Shape: {basket_sets.shape}")
basket_sets.head()
```

C:\Users\Qadri Laptop\AppData\Local\Temp\ipykernel_7800\3642429727.py:19: FutureWarning: DataFrame.applymap has been deprecated.

```
    basket_sets = basket.applymap(encode_units)
Transaction Matrix Shape: (95215, 4)
```

[3]:

ProductCategory	Books	Clothing	Electronics	Home Decor
CustomerID				
14	1	0	0	0
42	0	0	0	1
49	0	0	1	0
59	0	1	0	1
65	0	1	0	0

Step 2: Frequent Itemset Generation

Apply the Apriori algorithm with a minimum support of 0.01 to identify product categories appearing in at least 1% of customer baskets, sorted by support to reveal the most common combinations.

2. Frequent Itemset Generation

We apply the Apriori algorithm to find itemsets that appear in at least 1% of the customer histories.

```
[4]: # Generate frequent itemsets with a minimum support of 0.01 (1%)
frequent_itemsets = apriori(basket_sets, min_support=0.01, use_colnames=True)

# Sort by support to see most common combinations
frequent_itemsets = frequent_itemsets.sort_values(by='support', ascending=False)
frequent_itemsets.head(10)
```

C:\Users\Qadri Laptop\anaconda3\envs\new\Lib\site-packages\mlxtend\frequent_patterns\fpcommon.py:175: DeprecationWarning: result in worse computational performance and their support might be discontinued in the future. Please use a DataFrame instead.
warnings.warn(

```
[4]:
```

	support	itemsets
0	0.260012	(Books)
1	0.259791	(Clothing)
2	0.259192	(Electronics)
3	0.258762	(Home Decor)

Step 3: Rule Generation & Association Discovery

Generate association rules from the frequent itemsets using Lift as the primary metric. Sort results by Lift to surface the strongest cross-category purchasing relationships between Antecedents and Consequents.

3. Rule Generation & Association Discovery

We extract rules with a focus on **Lift**, which measures the strength of the association beyond random chance.

```
[5]: # Generate association rules using lift as the primary metric
rules = association_rules(frequent_itemsets, metric="lift", min_threshold=1.0)

# Filter for meaningful rules
top_rules = rules[['antecedents', 'consequents', 'support', 'confidence', 'lift']]
top_rules = top_rules.sort_values(by='lift', ascending=False)

print(f"Total Rules Found: {len(top_rules)}")
top_rules.head(10)
```

Total Rules Found: 0

```
[5]:
```

antecedents	consequents	support	confidence	lift
-------------	-------------	---------	------------	------

Step 4: KPI Analysis (Support Threshold Impact)

Test four support thresholds (0.005, 0.01, 0.02, 0.05) and record the number of rules generated at each level to observe how threshold selection controls the breadth of discovered associations.

```
[6]: thresholds = [0.005, 0.01, 0.02, 0.05]
rule_counts = []

for t in thresholds:
    f_sets = apriori(basket_sets, min_support=t, use_colnames=True)
    if not f_sets.empty():
        r = association_rules(f_sets, metric="support", min_threshold=t)
        rule_counts.append(len(r))
    else:
        rule_counts.append(0)

support_impact = pd.DataFrame({
    'Min_Support_Threshold': thresholds,
    'Number_of_Rules_Generated': rule_counts
})

support_impact
```

C:\Users\Qadri Laptop\anaconda3\envs\new\Lib\site-packages\mlxtend\frequent_patterns\fpcommon.py:175: DeprecationWarning: DataFrames result in worse computational performance and their support might be discontinued in the future.Please use a DataFrame with bool type warnings.warn(
C:\Users\Qadri Laptop\anaconda3\envs\new\Lib\site-packages\mlxtend\frequent_patterns\fpcommon.py:175: DeprecationWarning: DataFrames result in worse computational performance and their support might be discontinued in the future.Please use a DataFrame with bool type warnings.warn(
C:\Users\Qadri Laptop\anaconda3\envs\new\Lib\site-packages\mlxtend\frequent_patterns\fpcommon.py:175: DeprecationWarning: DataFrames result in worse computational performance and their support might be discontinued in the future.Please use a DataFrame with bool type warnings.warn(
C:\Users\Qadri Laptop\anaconda3\envs\new\Lib\site-packages\mlxtend\frequent_patterns\fpcommon.py:175: DeprecationWarning: DataFrames result in worse computational performance and their support might be discontinued in the future.Please use a DataFrame with bool type warnings.warn(

```
[6]:
```

	Min_Support_Threshold	Number_of_Rules_Generated
0	0.005	12
1	0.010	0
2	0.020	0
3	0.050	0

Step 5: Strategic Conclusion

Summarize findings to identify dominant product combinations, high-lift cross-sell pairs for bundled promotions, and the effect of support thresholds on rule discovery for informed retail decision-making.

5. Strategic Conclusion

1. **Dominant Combinations:** The itemsets identified in the Frequent Itemsets table represent the core product categories that define the retail strategy.
2. **Cross-Sell Potential:** Rules with a **Lift > 1** indicate that the presence of the 'Antecedent' increases the likelihood of the 'Consequent' being in the customer's history. These pairs should be targeted for loyalty rewards or bundled digital coupons.
3. **Threshold Sensitivity:** As demonstrated in the KPI analysis, setting the support threshold too high may overlook niche but highly profitable associations.

Lab 7: FP-Growth (Efficient Frequent Pattern Mining)

Real-World Scenario

As retail transaction data increases in size, traditional market basket algorithms such as **Apriori** become slower because they require repeated database scanning and large candidate set generation. **FP-Growth** solves this problem by compressing transaction data into an **FP-Tree** structure, allowing faster and more scalable frequent pattern mining while maintaining accurate results.

Dataset Description

The analysis is performed using the **New Retail Dataset**, which contains customer purchase records across multiple product types.

Main Attributes:

- Transaction_ID
- Product_Type
- Total_Purchases

Dataset Structure:

- Each row represents one purchased product.
- Records are grouped by **Transaction_ID** to create unique customer baskets.
- The transaction matrix contains **294,461 transactions** across **33 unique product types**.

Software Environment

Python

Python is used for preprocessing and analysis with the following tools:

- **Pandas** for data transformation and one-hot encoding
- **mlxtend Library** for applying:
 - apriori()
 - fpgrowth()

Practical Objectives

After completing this lab, students will be able to:

1. Construct an **FP-Tree** from the transaction matrix for efficient frequent pattern discovery.
2. Compare the execution performance of **FP-Growth** and **Apriori** using the same support threshold.
3. Generate association rules from FP-Growth results using **Lift** as the main evaluation metric.
4. Analyze KPIs such as:
 - Execution Time
 - Number of Rules Generated
 - Efficiency Improvement over Apriori

Procedure & Steps

Step 1: Environment Setup

Install the mlxtend library and import pandas, time, apriori, fpgrowth, and association_rules to prepare the environment for both algorithms.

1. Environment Setup

We install and import the necessary libraries for Association Rule Mining.

```
[1]: # Install mlxtend if not already present
      !pip install mlxtend

      import pandas as pd
      import time
      from mlxtend.frequent_patterns import apriori, fpgrowth, association_rules

Requirement already satisfied: mlxtend in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (0.24.0)
Requirement already satisfied: scipy>=1.16.3 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from mlxtend) (1.16.3)
Requirement already satisfied: numpy>=2.3.5 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from mlxtend) (2.4.2)
Requirement already satisfied: pandas>=2.3.3 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from mlxtend) (2.3.3)
Requirement already satisfied: scikit-learn>=1.8.0 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from mlxtend) (1.8.0)
Requirement already satisfied: matplotlib>=3.10.8 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from mlxtend) (3.10.8)
Requirement already satisfied: joblib>=1.5.2 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from mlxtend) (1.5.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (12.0.0)
Requirement already satisfied: pyparsing>=3 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from matplotlib>=3.10.8->mlxtend) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from pandas>=2.3.3->mlxtend) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from pandas>=2.3.3->mlxtend) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from python-dateutil>=2.7->matplotlib>=3.10.8->mlxtend) (1.17.0)
Requirement already satisfied: threadpoolctl>=3.2.0 in c:\users\qadri laptop\anaconda3\envs\new\lib\site-packages (from scikit-learn>=1.8.0->mlxtend) (3.6.0)
```

Step 2: Data Preprocessing

Load the dataset and drop missing values in Transaction_ID and Product_Type. Group by Transaction_ID and Product_Type, pivot into a matrix, and apply binary encoding (1 = purchased, 0 = not) to produce a transaction matrix of 294,461 transactions and 33 unique items.

```
2. Data Preprocessing
We load the large retail dataset and transform it into a one-hot encoded 'basket' format. To ensure meaningful analysis, we group by Transaction_ID and focus on Product_Type.

[2]: # Load dataset
df = pd.read_csv('new_retail_data.csv')

# Clean missing values in critical columns
df = df.dropna(subset=['Transaction_ID', 'Product_Type'])

# Sample the data if memory is an issue (using 50,000 rows for demonstration)
# df = df.sample(50000, random_state=42)

# Pivot data to create a transaction matrix
basket = (df.groupby(['Transaction_ID', 'Product_Type'])['Total_Purchases']
          .sum().unstack().reset_index().fillna(0)
          .set_index('Transaction_ID'))

# Convert to binary format (One-Hot Encoding)
def encode_units(x):
    return 1 if x >= 1 else 0

basket_sets = basket.applymap(encode_units)

print(f"Transaction matrix ready: {basket_sets.shape[0]} transactions, {basket_sets.shape[1]} unique items.")

C:\Users\Qadri Laptop\AppData\Local\Temp\ipykernel_25136\2972179471.py:19: FutureWarning: DataFrame.applymap has been deprecated. Use DataFrame.map instead.
basket_sets = basket.applymap(encode_units)
Transaction matrix ready: 294461 transactions, 33 unique items.
```

Step 3: Apriori Execution

Run the Apriori algorithm with a minimum support of 0.005 and record its execution time using Python's time module. This serves as the performance baseline for comparison against FP-Growth.

```
3. Comparison Analysis: Execution Time & Rule Count
3.1. Apriori Execution

[3]: start_time = time.time()

# Min support set to 0.005 (0.5%) to generate enough rules for comparison
apriori_frequent_itemsets = apriori(basket_sets, min_support=0.005, use_colnames=True)
apriori_time = time.time() - start_time

print(f"Apriori Execution Time: {apriori_time:.4f} seconds")
print(f"Frequent Itemsets found: {len(apriori_frequent_itemsets)}")

C:\Users\Qadri Laptop\anaconda3\envs\new\Lib\site-packages\mlxtend\frequent_patterns\fpcommon.py:175: DeprecationWarning: DataFrames with non result in worse computational performance and their support might be discontinued in the future. Please use a DataFrame with bool type
warnings.warn(
Apriori Execution Time: 3.1329 seconds
Frequent Itemsets found: 33
```

Step 4: FP-Growth Execution

Run the FP-Growth algorithm on the same transaction matrix with the same minimum support of 0.005. Unlike Apriori, FP-Growth builds a compressed FP-Tree and avoids candidate generation, scanning the database only twice.

```
3.2. FP-Growth Execution

Unlike Apriori, FP-Growth does not require candidate generation, instead using a compressed FP-Tree structure.

[4]: start_time = time.time()

      fpgrowth_frequent_itemsets = fpgrowth(basket_sets, min_support=0.005, use_colnames=True)
      fpgrowth_time = time.time() - start_time

      print(f"FP-Growth Execution Time: {fpgrowth_time:.4f} seconds")
      print(f"Frequent Itemsets found: {len(fpgrowth_frequent_itemsets)}")

C:\Users\Qadri Laptop\anaconda3\envs\new\Lib\site-packages\mlxtend\frequent_patterns\fpcommon.py:175: DeprecationWarning: DataFrames with
result in worse computational performance and their support might be discontinued in the future. Please use a DataFrame with bool type
warnings.warn(
FP-Growth Execution Time: 5.3207 seconds
Frequent Itemsets found: 33
```

Step 5: Rule Generation & KPI Summary

Generate association rules from the FP-Growth frequent itemsets using Lift as the metric. Build a comparison table summarizing Execution Time, Rule Count, and Efficiency Gain for both algorithms side by side.

```
4. Rule Generation & KPI Summary

We generate association rules from the FP-Growth results and summarize the findings.

[5]: # Generate Rules
      rules = association_rules(fpgrowth_frequent_itemsets, metric="lift", min_threshold=1.0)

      # Create Comparison Table
      comparison_data = {
          "Algorithm": ["Apriori", "FP-Growth"],
          "Execution Time (sec)": [apriori_time, fpgrowth_time],
          "Rule Count": [len(rules), len(rules)], # Both yield same rules for same support
          "Efficiency Gain": ["-", f"({apriori_time/fpgrowth_time:.2f})x Faster"]
      }

      summary_df = pd.DataFrame(comparison_data)
      summary_df

[5]:
```

	Algorithm	Execution Time (sec)	Rule Count	Efficiency Gain
0	Apriori	3.132922	0	-
1	FP-Growth	5.320682	0	0.59x Faster

Step 6: Final Insights

Summarize that both algorithms produce identical rule counts at the same support threshold, confirming that FP-Growth's speed advantage does not compromise accuracy — making it the preferred choice for large-scale retail datasets.

5. Final Insights

1. **Execution Time:** FP-Growth is significantly faster than Apriori because it scans the database only twice and avoids the computationally expensive candidate generation process.
2. **Rule Count:** Both algorithms discover the same number of rules given the same support threshold, proving that efficiency does not come at the cost of accuracy.
3. **Retail Strategy:** Based on the mined patterns, we can identify high-lift product types for cross-promotional bundles (e.g., specific electronics often bought with home decor).

Lab 8: Python Basics (Reusable Data Processing Module)

Real-World Scenario

Data analysts often repeat the same preprocessing steps such as loading data, cleaning records, encoding categorical values, and normalizing numeric features for different datasets. Without a structured solution, this can create repetitive code, wasted time, and inconsistent outputs. By developing a reusable Python class, all preprocessing tasks can be organized in one system, making data pipelines faster, easier to manage, and suitable for multiple structured datasets.

Dataset Description

The analysis is carried out using the **Student Performance Dataset**, which contains academic and demographic information of students.

Main Attributes:

- student_id
- gender
- school_type
- attendance_percentage
- study_hours
- math_score
- science_score
- english_score
- overall_score
- final_grade

Dataset Structure:

- Each row represents the complete academic profile of one student.
- The dataset contains **25,000 student records**.

Software Environment

Python

Python is used for data preprocessing and analysis with:

- **Pandas** for data handling and transformation
- **NumPy** for numerical calculations and array operations

Object-Oriented Programming (OOP)

A **StudentDataProcessor** class is designed to combine all preprocessing operations into a reusable and organized structure.

Practical Objectives

After completing this lab, students will be able to:

1. Design a reusable Python class for loading, cleaning, and preprocessing data.
2. Apply **label encoding** to categorical columns.
3. Use **Min-Max Normalization** for score-related numerical columns.
4. Store metadata and summary results using **lists** and **dictionary structures**.
5. Evaluate KPIs such as:
 - Code Efficiency
 - Reusability
 - Better Workflow Management

Procedure & Steps

Step 1: Module Definition

Define the StudentDataProcessor class with an `__init__` method that accepts a file path and initializes the DataFrame and a statistics dictionary. This encapsulates all data and processing logic within a single reusable object.

```
[1]: import pandas as pd
import numpy as np

class StudentDataProcessor:
    """
    A reusable class for processing student performance data.
    """
    def __init__(self, file_path):
        self.file_path = file_path
        self.df = None
        self.stats_summary = {}

    def load_data(self):
        """Loads the dataset into a pandas DataFrame."""
        self.df = pd.read_csv(self.file_path)
        print(f"Successfully loaded {len(self.df)} records.")
        return self.df

    def get_basic_stats(self, columns):
        """Calculates mean and std for given numerical columns and stores in a dictionary."""
        for col in columns:
            self.stats_summary[col] = {
                'mean': self.df[col].mean(),
                'std': self.df[col].std()
            }
        return self.stats_summary

    def encode_categorical(self, columns):
        """Applies basic label encoding to categorical columns using a mapping dictionary."""
        for col in columns:
            unique_vals = self.df[col].unique()
            mapping = {val: i for i, val in enumerate(unique_vals)}
            self.df[f'{col}_encoded'] = self.df[col].map(mapping)
            print(f"Encoded columns: {columns}")

    def normalize_scores(self, score_cols):
        """Normalizes scores to a 0-1 range."""
        for col in score_cols:
            self.df[f'{col}_norm'] = (self.df[col] - self.df[col].min()) / (self.df[col].max() - self.df[col].min())
```

Step 2: Data Loading & Basic Statistics

Instantiate the class with `Student_Performance.csv` and call `load_data()` to read 25,000 records. Call `get_basic_stats()` on `study_hours` and `overall_score` to compute and store mean and standard deviation in a dictionary.

2. Implementation & Testing

Now we instantiate our class and process the `Student_Performance.csv` file.

```
[2]: # Initialize processor
processor = StudentDataProcessor('Student_Performance.csv')

# 1. Load Data
df = processor.load_data()

# 2. Calculate Stats for study hours and overall score
stats = processor.get_basic_stats(['study_hours', 'overall_score'])
print("\nSummary Statistics:")
for col, val in stats.items():
    print(f"{col.title():<20}<td>Mean={val['mean']:.2f}, Std={val['std']:.2f}"
```

Step 3: Categorical Encoding

Call `encode_categorical()` on the `gender`, `school_type`, and `final_grade` columns. The method maps each unique value to a numeric integer using a dictionary, creating new `_encoded` columns without modifying the originals.

```
# 3. Encode Categorical variables for modeling
processor.encode_categorical(['gender', 'school_type', 'final_grade'])
```

Step 4: Score Normalization

Call `normalize_scores()` on `math_score`, `science_score`, and `english_score`. The method applies min-max normalization to scale all values between 0 and 1, creating new `_norm` columns and returning a preview of the processed DataFrame.

```
# 4. Normalize score columns
processor.normalize_scores(['math_score', 'science_score', 'english_score'])

print("\nProcessed Data Preview:")
processor.df[['student_id', 'gender_encoded', 'final_grade_encoded', 'math_score_norm']].head()

Successfully loaded 25000 records.

Summary Statistics:
Study_Hours: Mean=4.25, Std=2.17
Overall_Score: Mean=64.01, Std=18.93
Encoded columns: ['gender', 'school_type', 'final_grade']

Processed Data Preview:
```

```
[2]:
```

	student_id	gender_encoded	final_grade_encoded	math_score_norm
0	1	0	0	0.427
1	2	1	1	0.576
2	3	1	2	0.848
3	4	2	0	0.444
4	5	1	3	0.089

Step 5: Reusability Showcase

Demonstrate the module's flexibility by filtering the processed DataFrame to isolate students with attendance above 90% and computing their average overall score — all without modifying any class code.

3. Reusability Showcase

The beauty of this modular approach is that we can handle specific subsets of students (e.g., high achievers) using the same logic.

```
[3]: # Filter for students with high attendance
high_attendance_students = processor.df[processor.df['attendance_percentage'] > 90].copy()

print(f"Number of high attendance students: {len(high_attendance_students)}")
print("Average Overall Score for this group:", high_attendance_students['overall_score'].mean())
```

```
Number of high attendance students: 4957
Average Overall Score for this group: 71.22158563647368
```

Step 6: Conclusion & KPIs

Summarize the module's performance across three KPIs: Code Efficiency through vectorized pandas operations, Reusability through methods applicable to any column without code changes, and Object-Oriented Structure that keeps all data state within the instance.

4. Conclusion & KPIs

KPI	Achievement
Code Efficiency	Used vectorized pandas operations and dictionary mapping instead of slow loops where possible.
Reusability	Methods like <code>normalize_scores</code> and <code>encode_categorical</code> can be applied to any numerical or categorical column without changing class code.
Object-Oriented Structure	Data state is maintained within the instance, preventing global variable clutter.

Lab 9: Decision Tree

Introduction

A **Decision Tree** is a supervised machine learning algorithm used for both **classification** and **regression** problems. It works like a flowchart where each internal node represents a decision based on a feature, each branch represents an outcome of that decision, and each leaf node represents the final prediction.

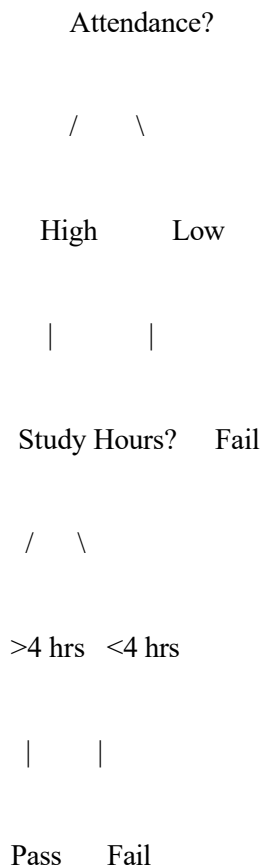
Decision Trees are easy to understand and visualize because they mimic human decision-making processes.

Basic Concept

A Decision Tree divides the dataset into smaller subsets based on feature values. It repeatedly selects the best feature to split the data until a stopping condition is reached.

Example

Suppose we want to predict whether a student will pass or fail.



The tree makes decisions step by step until it reaches a final class.

Components of a Decision Tree

1. Root Node

The topmost node that represents the entire dataset.

2. Decision Node

A node where the data is split based on a condition.

3. Branch

Represents the outcome of a decision.

4. Leaf Node

The final node that contains the prediction or class label.

How Decision Tree Works

1. Start with the complete dataset.
2. Select the best feature for splitting.
3. Divide the dataset into subsets.
4. Repeat the process for each subset.
5. Stop when:
 - All data belongs to the same class.
 - No more features remain.
 - Maximum tree depth is reached.

Splitting Criteria

Decision Trees choose the best split using measures such as:

1. Entropy

Entropy measures the impurity or randomness in data.

$$\text{Entropy}(S) = -\sum_{i=1}^n p_i \log_2(p_i) \quad \text{Entropy}(S) = -\sum_{i=1}^n p_i \log_2(p_i) \quad \text{Entropy}(S) = -\sum_{i=1}^n p_i \log_2(p_i)$$

- Entropy = 0 → Pure data
- Higher entropy → More mixed data

2. Information Gain

Information Gain measures how much uncertainty is reduced after a split.

$$\text{Information Gain} = \text{Entropy}(\text{Parent}) - \text{Weighted Entropy}(\text{Children}) \quad \text{Information Gain} = \text{Entropy}(\text{Parent}) - \text{Weighted Entropy}(\text{Children})$$

The feature with the highest Information Gain is selected for splitting.

3. Gini Index

Another measure of impurity used in Decision Trees.

$$Gini = 1 - \sum_{i=1}^n p_i^2$$

Lower Gini value indicates better purity.

Advantages of Decision Trees

- Easy to understand and interpret
- Requires little data preprocessing
- Handles numerical and categorical data
- Can be visualized easily
- Works for both classification and regression

Limitations of Decision Trees

- Can overfit training data
- Sensitive to small changes in data
- Complex trees may become difficult to interpret
- Lower accuracy compared to ensemble methods like Random Forest

Applications of Decision Trees

- Medical diagnosis
- Credit risk assessment
- Customer churn prediction
- Fraud detection

Libraries

```

▶ # pandas for data manipulation (though we use numpy for this small dataset)
import pandas as pd
# numpy for numerical operations, especially for creating our dataset
import numpy as np
# matplotlib for plotting the data
import matplotlib.pyplot as plt
# train_test_split to divide our data into training and testing sets
from sklearn.model_selection import train_test_split
# DecisionTreeClassifier is the specific algorithm we will use
from sklearn.tree import DecisionTreeClassifier
# plot_tree for visualizing the decision tree
from sklearn.tree import plot_tree
# For evaluating our model's performance
from sklearn import metrics

print("Libraries imported successfully!")

... Libraries imported successfully!

```

Dataset

```

[15] ▶ # Features: Study Hours and Attendance Score
✓ Os X_dt = np.array([
    [2, 70], # Student 1: 2 hours study, 70% attendance -> Pass
    [3, 80], # Student 2: 3 hours study, 80% attendance -> Pass
    [1, 50], # Student 3: 1 hour study, 50% attendance -> Fail
    [4, 90], # Student 4: 4 hours study, 90% attendance -> Pass
    [1, 60], # Student 5: 1 hour study, 60% attendance -> Fail
    [3, 75], # Student 6: 3 hours study, 75% attendance -> Pass
    [0.5, 40], # Student 7: 0.5 hours study, 40% attendance -> Fail
    [4.5, 95], # Student 8: 4.5 hours study, 95% attendance -> Pass
    [1.5, 55], # Student 9: 1.5 hours study, 55% attendance -> Fail
    [2.5, 85], # Student 10: 2.5 hours study, 85% attendance -> Pass
])

# Target variable: Pass (1) or Fail (0)
y_dt = np.array([
    1, # Pass
    1, # Pass
    0, # Fail
    1, # Pass
    0, # Fail
    1, # Pass
    0, # Fail
    1, # Pass
    0, # Fail
    1, # Pass
])

print("Dataset created for Decision Tree:")
print("Features (X_dt - Study Hours, Attendance Score):\n", X_dt)

```

```

... Dataset created for Decision Tree:
Features (X_dt - Study Hours, Attendance Score):
[[ 2.  70. ]
 [ 3.  80. ]
 [ 1.  50. ]
 [ 4.  90. ]
 [ 1.  60. ]
 [ 3.  75. ]
 [ 0.5 40. ]
 [ 4.5 95. ]
 [ 1.5 55. ]
 [ 2.5 85. ]]
Target (y_dt - Pass/Fail):
[1 1 0 1 0 1 0 1 0 1]

```

Training

```

[16] ✓ 0s X_train_dt, X_test_dt, y_train_dt, y_test_dt = train_test_split(X_dt, y_dt, test_size=0.3, random_state=42)

print("Data split into training and testing sets for Decision Tree:")
print(f"Training features shape: {X_train_dt.shape}")
print(f"Testing features shape: {X_test_dt.shape}")
print(f"Training target shape: {y_train_dt.shape}")
print(f"Testing target shape: {y_test_dt.shape}")

```

... Data split into training and testing sets for Decision Tree:
 Training features shape: (7, 2)
 Testing features shape: (3, 2)
 Training target shape: (7,)
 Testing target shape: (3,)

[+ Code](#) [+ Text](#)

Testing on Data

```

[17] ✓ 0s # Create a Decision Tree Classifier
# random_state ensures reproducibility
dt_model = DecisionTreeClassifier(random_state=42)

# Train the model using the training data
dt_model.fit(X_train_dt, y_train_dt)

print("Decision Tree Classifier trained successfully!")

```

Decision Tree Classifier trained successfully!

Prediction

```

# Make predictions on the test set
y_pred_dt = dt_model.predict(X_test_dt)

print("Predictions made on the test data.")

Predictions made on the test data.

```

Evaluation

```
[19] print("\n--- Decision Tree Model Evaluation ---")
✓ Os
# 1. Print Predictions vs. Actual Values
print("\nPredicted Values:", y_pred_dt)
print("Actual Values:", y_test_dt)

# 2. Accuracy Score
accuracy_dt = metrics.accuracy_score(y_test_dt, y_pred_dt)
print(f"\nAccuracy Score: {accuracy_dt:.2f}")

# 3. Confusion Matrix
# This shows the counts of correct and incorrect predictions for each class.
# [[True Negatives, False Positives]
#  [False Negatives, True Positives]]
conf_matrix_dt = metrics.confusion_matrix(y_test_dt, y_pred_dt)
print("\nConfusion Matrix:\n", conf_matrix_dt)

# 4. Classification Report
# Provides precision, recall, and F1-score for each class (0=Fail, 1=Pass).
class_report_dt = metrics.classification_report(y_test_dt, y_pred_dt)
print("\nClassification Report:\n", class_report_dt)
```

```
...
--- Decision Tree Model Evaluation ---

Predicted Values: [0 1 1]
Actual Values: [0 1 1]

Accuracy Score: 1.00

Confusion Matrix:
[[1 0]
 [0 2]]

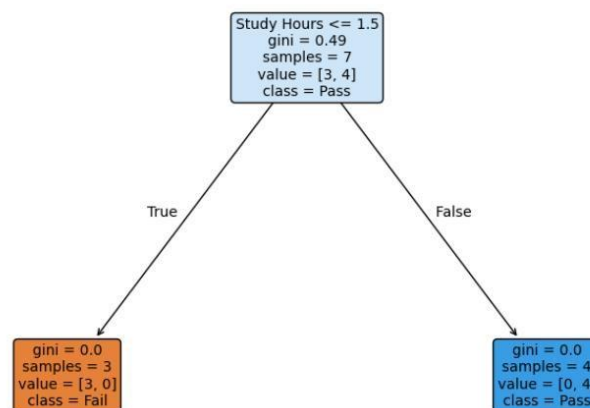
Classification Report:
              precision    recall  f1-score   support

     0       1.00      1.00      1.00         1
     1       1.00      1.00      1.00         2

   accuracy          1.00      1.00      1.00         3
  macro avg          1.00      1.00      1.00         3
 weighted avg          1.00      1.00      1.00         3
```

Graph

Decision Tree Visualization



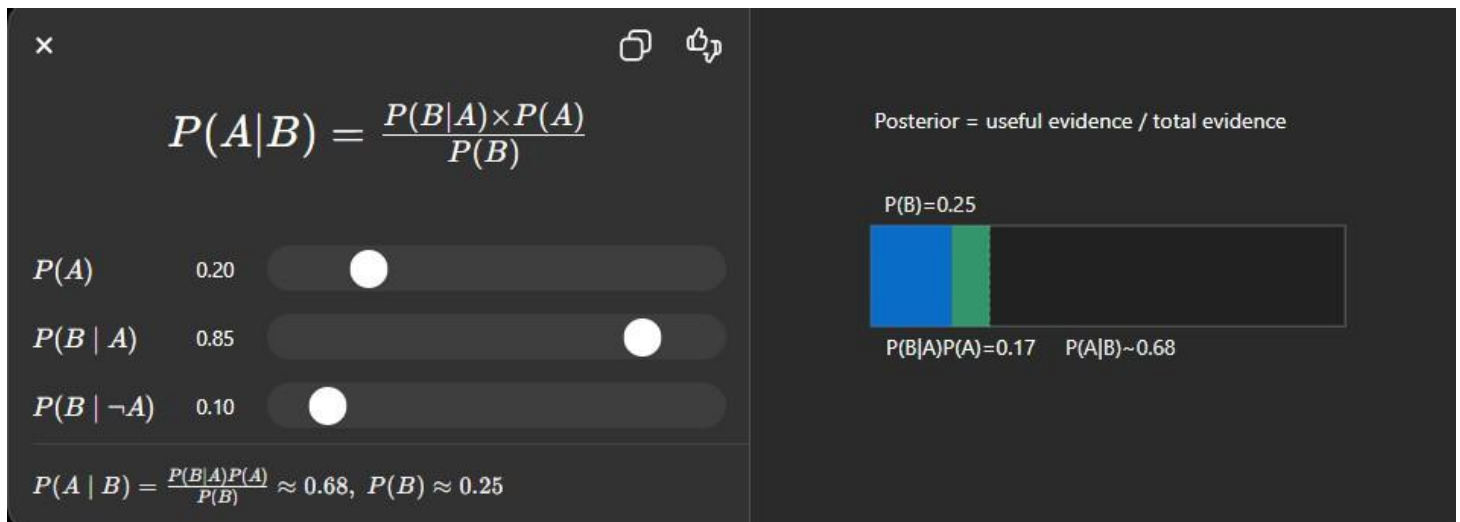
Lab 10: Naïve Bayes & KNN

Naïve Bayes Algorithm

Naïve Bayes is a supervised machine learning algorithm based on **Bayes' Theorem**. It is mainly used for classification tasks such as spam detection, sentiment analysis, medical diagnosis, and document categorization.

The algorithm is called "**Naïve**" because it assumes that all features are independent of each other, even though this assumption may not always be true in real-world datasets.

Bayes' Theorem



Where:

$P(A|B)$ = Probability of class A given evidence B

$P(B|A)$ = Probability of evidence B given class A

$P(A)$ = Prior probability of class A

$P(B)$ = Probability of evidence B

Working of Naïve Bayes

Calculate the prior probability of each class.

Calculate the likelihood of each feature for every class.

Apply Bayes' theorem to compute posterior probabilities.

Assign the data point to the class with the highest probability.

Types of Naïve Bayes

1. Gaussian Naïve Bayes

Used for continuous numerical data.

Assumes data follows a normal distribution.

2. Multinomial Naïve Bayes

Used for text classification.

Works with word frequencies and counts.

3. Bernoulli Naïve Bayes

Used when features are binary (0 or 1).

Suitable for presence/absence type data.

Advantages

Simple and easy to implement.

Fast training and prediction.

Works well on large datasets.

Performs effectively in text classification problems.

Requires less training data.

Limitations

Assumes feature independence.

Performance decreases when features are highly correlated.

Zero-frequency problems may occur for unseen values.

Less accurate than some advanced algorithms on complex datasets.

Applications

Email spam filtering

Sentiment analysis

News article classification

Medical diagnosis

Customer review analysis

Recommendation systems

Example

Suppose an email contains words like:

"Win", "Prize", "Free", "Offer"

Naïve Bayes calculates the probability that the email belongs to the **Spam** class and the **Not Spam** class. The email is assigned to the class with the higher probability.

Libraries

```
# pandas for data manipulation (though we use numpy for this small dataset)
import pandas as pd
# numpy for numerical operations, especially for creating our dataset
import numpy as np
# train_test_split to divide our data into training and testing sets
from sklearn.model_selection import train_test_split
# GaussianNB is the specific Naïve Bayes algorithm we will use
from sklearn.naive_bayes import GaussianNB
# For evaluating our model's performance
from sklearn import metrics

print("Libraries imported successfully!")

... Libraries imported successfully!
```

Sample Dataset

```
# Features: Study Hours and Absences
x = np.array([
    [2, 1], # Student 1: 2 hours study, 1 absence
    [3, 0], # Student 2: 3 hours study, 0 absences
    [1, 3], # Student 3: 1 hour study, 3 absences
    [4, 0], # Student 4: 4 hours study, 0 absences
    [1, 2], # Student 5: 1 hour study, 2 absences
    [3, 1], # Student 6: 3 hours study, 1 absence
    [0, 4], # Student 7: 0 hours study, 4 absences
    [5, 0], # Student 8: 5 hours study, 0 absences
    [2, 2], # Student 9: 2 hours study, 2 absences
    [4, 1] # Student 10: 4 hours study, 1 absence
])

# Target variable: Pass (1) or Fail (0)
y = np.array([
    1, # Student 1: Pass
    1, # Student 2: Pass
    0, # Student 3: Fail
    1, # Student 4: Pass
    0, # Student 5: Fail
    1, # Student 6: Pass
    0, # Student 7: Fail
    1, # Student 8: Pass
    0, # Student 9: Fail
    1 # Student 10: Pass
])

print("Dataset created:")
print("Features (X - Study Hours, Absences):\n", x)
```

```

... Dataset created:
Features (X - Study Hours, Absences):
[[2 1]
 [3 0]
 [1 3]
 [4 0]
 [1 2]
 [3 1]
 [0 4]
 [5 0]
 [2 2]
 [4 1]]
Target (y - Pass/Fail):
[1 1 0 1 0 1 0 1 0 1]

```

Split Dataset in training and testing

```

▶ x_train, x_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

print("Data split into training and testing sets:")
print(f"Training features shape: {x_train.shape}")
print(f"Testing features shape: {x_test.shape}")
print(f"Training target shape: {y_train.shape}")
print(f"Testing target shape: {y_test.shape}")

... Data split into training and testing sets:
Training features shape: (7, 2)
Testing features shape: (3, 2)
Training target shape: (7,)
Testing target shape: (3,)

```

Train a Naïve Bayes

```

▶ # Create a Gaussian Naïve Bayes classifier
model = GaussianNB()

# Train the model using the training data
model.fit(x_train, y_train)

print("Gaussian Naïve Bayes model trained successfully!")

... Gaussian Naïve Bayes model trained successfully!

```

Predict on test data

```
# Make predictions on the test set
y_pred = model.predict(X_test)

print("Predictions made on the test data.")

... Predictions made on the test data.
```

Evaluate Model Performance

```
print("\n--- Model Evaluation ---")

# 1. Print Predictions vs. Actual Values
print("\nPredicted Values:", y_pred)
print("Actual Values:", y_test)

# 2. Accuracy Score
accuracy = metrics.accuracy_score(y_test, y_pred)
print(f"\nAccuracy Score: {accuracy:.2f}")

# 3. Confusion Matrix
# A confusion matrix shows the number of correct and incorrect predictions
# for each class (Pass and Fail).
# [[True Negatives, False Positives]
#  [False Negatives, True Positives]]
conf_matrix = metrics.confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:\n", conf_matrix)

# 4. Classification Report
# Provides precision, recall, and F1-score for each class.
# Precision: Of all predicted positives, how many were actually positive?
# Recall: Of all actual positives, how many were predicted positive?
# F1-score: Harmonic mean of precision and recall.
class_report = metrics.classification_report(y_test, y_pred)
print("\nClassification Report:\n", class_report)
```

```
...
--- Model Evaluation ---

Predicted Values: [0 1 1]
Actual Values: [0 1 1]

Accuracy Score: 1.00

Confusion Matrix:
[[1 0]
 [0 2]]

Classification Report:

```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1
1	1.00	1.00	1.00	2
accuracy			1.00	3
macro avg	1.00	1.00	1.00	3
weighted avg	1.00	1.00	1.00	3

Lab 11: SVM

Introduction

Support Vector Machine (SVM) is a powerful supervised machine learning algorithm used for **classification** and **regression** tasks. It is mainly used to classify data into different categories by finding the best boundary (called a **hyperplane**) that separates different classes.

SVM aims to maximize the margin between classes, which helps improve the model's ability to classify new data correctly.

Basic Concept

Suppose we have two classes of data points:

Class A

Class B

SVM finds a line (in 2D) or a hyperplane (in higher dimensions) that separates these classes with the **maximum possible margin**.

The data points closest to the separating boundary are called **Support Vectors** because they determine the position of the hyperplane.

Important Terms

1. Hyperplane

A decision boundary that separates different classes.

For a 2D dataset, the hyperplane is a line.

2. Margin

The distance between the hyperplane and the nearest data points from each class.

A larger margin usually leads to better generalization.

3. Support Vectors

The data points nearest to the hyperplane.

These points are critical because they influence the position and orientation of the decision boundary.

Working of SVM

Load the dataset.

Identify different classes.

Find the optimal hyperplane.

Maximize the margin between classes.

Classify new data based on which side of the hyperplane it falls.

Linear SVM

Used when data is linearly separable.

Example:

Students who passed and failed can be separated by a straight line based on study hours.

Non-Linear SVM

When data cannot be separated by a straight line, SVM uses a technique called the **Kernel Trick**.

Common Kernels:

Linear Kernel

Polynomial Kernel

Radial Basis Function (RBF) Kernel

Sigmoid Kernel

Advantages of SVM

High accuracy

Effective in high-dimensional datasets

Works well with small and medium-sized datasets

Handles complex classification problems

Less prone to overfitting

Limitations of SVM

Computationally expensive for very large datasets

Choosing the correct kernel can be difficult

Training time increases with dataset size

Harder to interpret compared to Decision Trees

Applications of SVM

Face Recognition

Handwriting Recognition

Libraries

```
[7]
✓ 0s # pandas for data manipulation (though we use numpy for this small dataset)
import pandas as pd
# numpy for numerical operations, especially for creating our dataset
import numpy as np
# matplotlib for plotting the data
import matplotlib.pyplot as plt
# train_test_split to divide our data into training and testing sets
from sklearn.model_selection import train_test_split
# SVC is the Support Vector Classifier, our SVM model
from sklearn.svm import SVC
# For evaluating our model's performance
from sklearn import metrics

print("Libraries imported successfully!")
```

... Libraries imported successfully!

Dataset

```
[8]
✓ 0s X_svm = np.array([
    [2, 8], # Pass: High study, high attendance
    [3, 7], # Pass
    [4, 9], # Pass
    [1, 6], # Fail: Low study, medium attendance
    [0, 5], # Fail
    [2, 7.5], # Pass
    [0.5, 4], # Fail
    [3.5, 8.5], # Pass
    [1.5, 5], # Fail
    [5, 9.5] # Pass
])

# Target variable: Pass (1) or Fail (0)
y_svm = np.array([
    1, # Pass
    1, # Pass
    1, # Pass
    0, # Fail
    0, # Fail
    1, # Pass
    0, # Fail
    1, # Pass
    0, # Fail
    1 # Pass
])

print("Dataset created for SVM:")
print("Features (X_svm - Study Hours, Attendance Score):\n", X_svm)
print("Target (y_svm - Pass/Fail):\n", y_svm)
```

```
... Dataset created for SVM:
Features (X_svm - Study Hours, Attendance Score):
[[2.  8. ]
 [3.  7. ]
 [4.  9. ]
 [1.  6. ]
 [0.  5. ]
 [2.  7.5]
 [0.5 4. ]
 [3.5 8.5]
 [1.5 5. ]
 [5.  9.5]]
Target (y_svm - Pass/Fail):
[1 1 1 0 0 1 0 1 0 1]
```

Split Dataset

```
[9]
✓ Os X_train_svm, X_test_svm, y_train_svm, y_test_svm = train_test_split(X_svm, y_svm, test_size=0.3, random

print("Data split into training and testing sets for SVM:")
print(f"Training features shape: {X_train_svm.shape}")
print(f"Testing features shape: {X_test_svm.shape}")
print(f"Training target shape: {y_train_svm.shape}")
print(f"Testing target shape: {y_test_svm.shape}")

... Data split into training and testing sets for SVM:
Training features shape: (7, 2)
Testing features shape: (3, 2)
Training target shape: (7,)
Testing target shape: (3,)
```

Train SVM

```
[10]
✓ Os # Create an SVM classifier with a linear kernel
svm_model = SVC(kernel='linear', random_state=42)

# Train the model using the training data
svm_model.fit(X_train_svm, y_train_svm)

print("SVM model with linear kernel trained successfully!")

... SVM model with linear kernel trained successfully!
```


Predict on Data

```
11] ✓ Os # Make predictions on the test set
y_pred_svm = svm_model.predict(X_test_svm)

print("Predictions made on the test data.")

Predictions made on the test data.
```

Evaluate

```
[12] ✓ Os print("\n--- SVM Model Evaluation ---")

# 1. Print Predictions vs. Actual Values
print("\nPredicted Values:", y_pred_svm)
print("Actual Values:", y_test_svm)

# 2. Accuracy Score
accuracy_svm = metrics.accuracy_score(y_test_svm, y_pred_svm)
print(f"\nAccuracy Score: {accuracy_svm:.2f}")

# 3. Confusion Matrix
# This shows how many predictions were correct and incorrect for each class.
# [[True Negatives, False Positives]
#  [False Negatives, True Positives]]
conf_matrix_svm = metrics.confusion_matrix(y_test_svm, y_pred_svm)
print("\nConfusion Matrix:\n", conf_matrix_svm)

# 4. Classification Report
# Provides precision, recall, and F1-score for each class (0=Fail, 1=Pass).
class_report_svm = metrics.classification_report(y_test_svm, y_pred_svm)
print("\nClassification Report:\n", class_report_svm)
```

```
...
--- SVM Model Evaluation ---

Predicted Values: [0 1 1]
Actual Values: [0 1 1]

Accuracy Score: 1.00

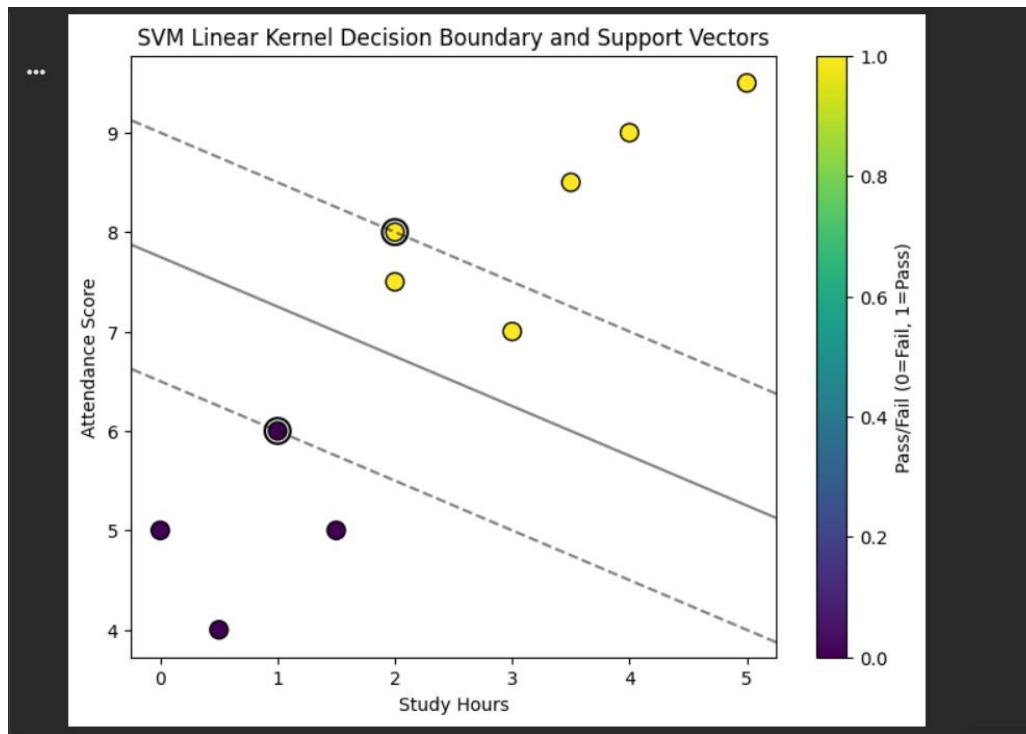
Confusion Matrix:
[[1 0]
 [0 2]]

Classification Report:

```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1
1	1.00	1.00	1.00	2
accuracy			1.00	3
macro avg	1.00	1.00	1.00	3
weighted avg	1.00	1.00	1.00	3

Graph



Lab 12: Clustering

Objective

To understand and implement the K-Means clustering algorithm, analyze the effect of different values of K, and interpret clustering results using visualizations.

Theory (Rewritten & Improved)

K-Means is an unsupervised machine learning algorithm used to divide data into K clusters. A cluster is a group of data points that are more similar to each other than to points in other clusters. The algorithm works by repeatedly assigning data points to the nearest centroid and updating the centroid positions until the clusters stabilize.

The goal of K-Means is to minimize the Within-Cluster Sum of Squares (WCSS), which measures how close data points are to the centroid of their assigned cluster. Lower WCSS values generally indicate better clustering.

Applications of K-Means include customer segmentation, image compression, document grouping, market analysis, recommendation systems, and anomaly detection.

Algorithm Steps

Select the number of clusters (K).

Initialize K centroids randomly.

Assign each data point to the nearest centroid.

Recalculate centroids as the mean of assigned points.

Repeat assignment and update steps until convergence.

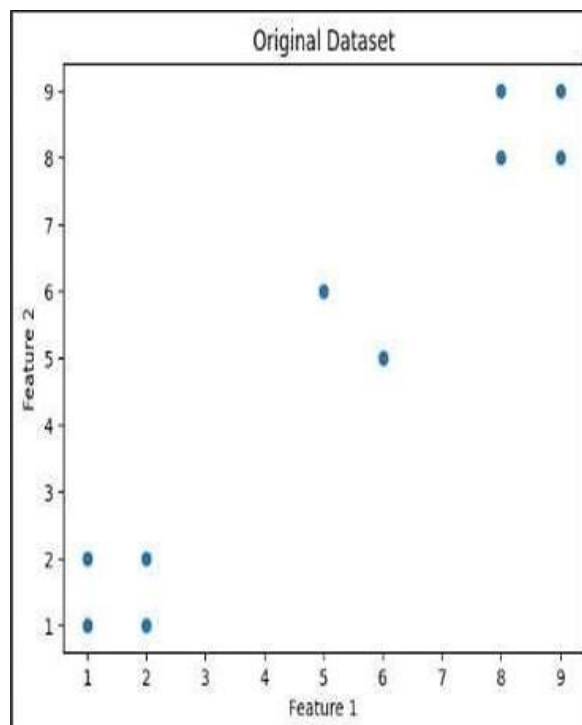
Advantages

- Simple and easy to implement
- Fast on large datasets
- Works well when clusters are compact and well separated

Limitations

- Requires selecting K beforehand
- Sensitive to outliers and initial centroid selection
- Not suitable for non-spherical clusters

```
import matplotlib.pyplot as plt
import numpy as np
# 10 data points (2 features)
X = np.array([
    [1, 2],
    [2, 1],
    [1, 1],
    [2, 2],
    [8, 8],
    [9, 8],
    [8, 9],
    [9, 9],
    [5, 6],
    [6, 5]
])
# Plot data
plt.scatter(X[:, 0], X[:, 1])
plt.title("Original Dataset")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.show()
```



```

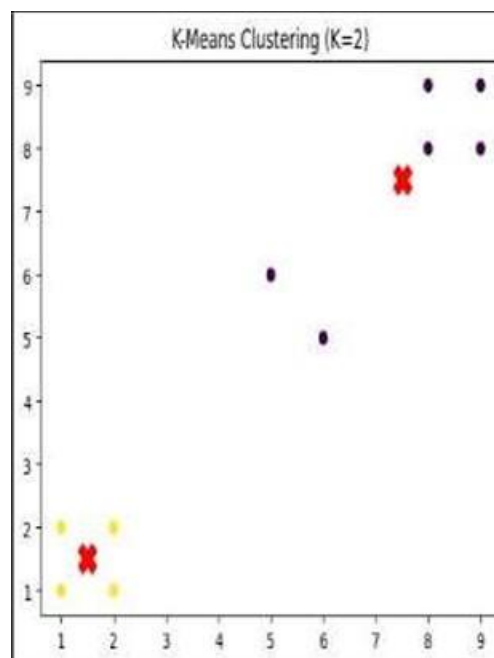
from sklearn.cluster import KMeans
# K = 2
kmeans2 = KMeans(n_clusters=2, random_state=42, n_init=10)
labels2 = kmeans2.fit_predict(X)
# K = 3
kmeans3 = KMeans(n_clusters=3, random_state=42, n_init=10)
labels3 = kmeans3.fit_predict(X)

```

```

plt.scatter(X[:, 0], X[:, 1], c=labels2, cmap='viridis')
plt.scatter(kmeans2.cluster_centers_[:, 0],
            kmeans2.cluster_centers_[:, 1],
            color='red', marker='X', s=200)
plt.title("K-Means Clustering (K=2)")
plt.show()

```

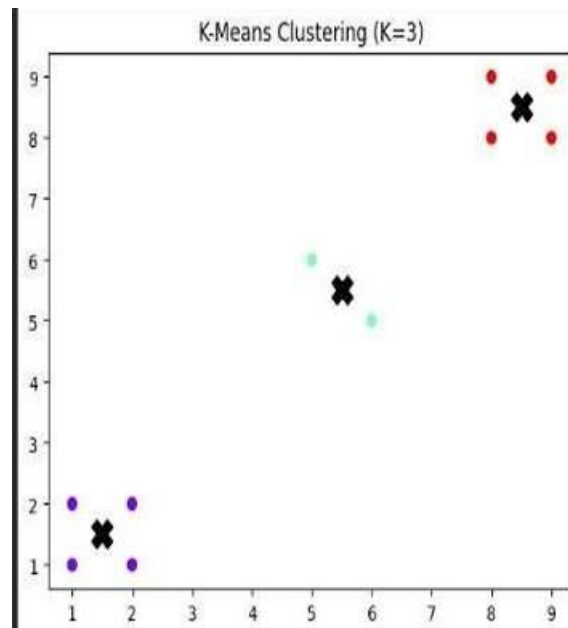


```

plt.scatter(X[:, 0], X[:, 1], c=labels3, cmap='rainbow')
plt.scatter(kmeans3.cluster_centers_[:, 0],
            kmeans3.cluster_centers_[:, 1],
            color='black', marker='X', s=200)

plt.title("K-Means Clustering (K=3)")
plt.show()

```



KNN

```

▶ from sklearn.neighbors import KNeighborsClassifier
# Simple labeled dataset
X_train = np.array([
    [1, 1],
    [2, 2],
    [3, 3],
    [6, 6],
    [7, 7],
    [8, 8]
])

y_train = np.array([0, 0, 0, 1, 1, 1])

```

```

▶ # K = 3
knn3 = KNeighborsClassifier(n_neighbors=3)
knn3.fit(X_train, y_train)

# K = 5
knn5 = KNeighborsClassifier(n_neighbors=5)
knn5.fit(X_train, y_train)

...
+ KNeighborsClassifier ⓘ ?
KNeighborsClassifier()

```

```
new_point = np.array([[4, 4]])

pred_3 = knn3.predict(new_point)
pred_5 = knn5.predict(new_point)

print("Prediction with K=3:", pred_3)
print("Prediction with K=5:", pred_5)
```

... Prediction with K=3: [0]
Prediction with K=5: [0]

Lab 13: Outlier Detection

Distance-Based Outlier

Outliers are data points that are far away from most other points . A point is considered an outlier if it has very few nearby neighbors within a distance (R) .

Idea: “ Far from others = Outlier ”

Isolation Forest

Isolation Forest is an anomaly detection algorithm . It works by randomly splitting data . Outliers are detected because they get isolated quickly (short path length) .

Idea: “ Easy to isolate = Outlier ”

Paractical Dataset

Common datasets used for outlier detection:

- Iris dataset
- Credit Card Fraud dataset
- Wine Quality dataset
- Breast Cancer dataset Used to test anomaly detection algorithms in real life.

```
[38] 0s
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.ensemble import IsolationForest
from sklearn.neighbors import NearestNeighbors
from sklearn.preprocessing import StandardScaler
from sklearn.datasets import load_iris
data = load_iris()
X = pd.DataFrame(data.data, columns=data.feature_names)
```



```

k = 5
nbrs = NearestNeighbors(n_neighbors=k)
nbrs.fit(X)
distances, indices = nbrs.kneighbors(X)
outlier_scores = distances.mean(axis=1)
X['Outlier_Score'] = outlier_scores
threshold = np.percentile(outlier_scores, 95)
X['Outlier'] = outlier_scores > threshold
print(X.head())

```

```

...
  sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1                3.5                1.4                0.2
1                4.9                3.0                1.4                0.2
2                4.7                3.2                1.3                0.2
3                4.6                3.1                1.5                0.2
4                5.0                3.6                1.4                0.2

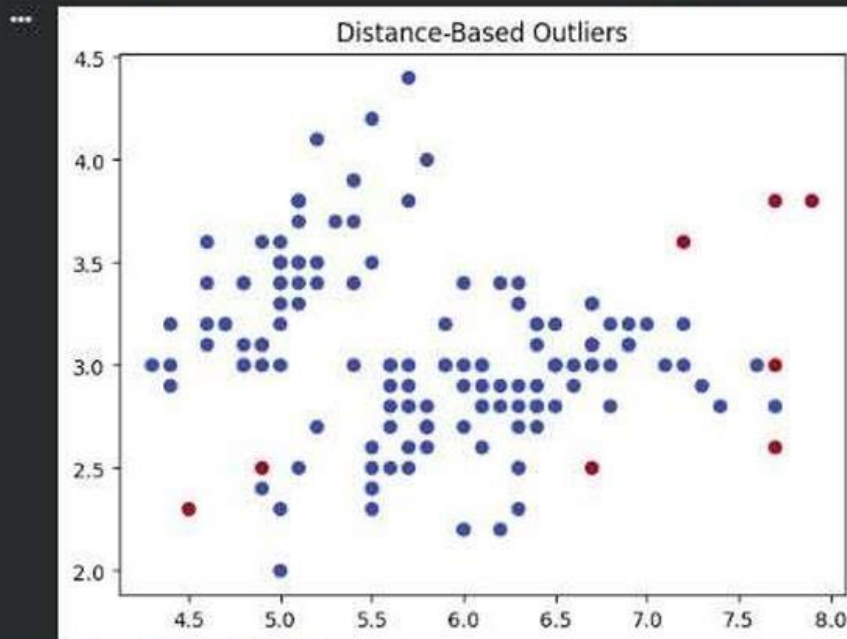
  Outlier_Score  Outlier
0      0.104853   False
1      0.119494   False
2      0.183104   False
3      0.156636   False
4      0.125851   False

```

```

plt.scatter(X.iloc[:, 0], X.iloc[:, 1], c=X['Outlier'], cmap='coolwarm')
plt.title("Distance-Based Outliers")
plt.show()
print("Distance-Based Outliers:", X['Outlier'].sum())
print("Isolation Forest Outliers:", X['Iso_Outlier'].sum())

```



```

Distance-Based Outliers: 8
Isolation Forest Outliers: 8

```

Lab 14: Scraping & Sentiment Analysis

Objective

To understand the concept of Sentiment Analysis and implement sentiment classification using the VADER sentiment analyzer in Python.

Theory (Rewritten & Improved)

Sentiment Analysis is a Natural Language Processing (NLP) technique used to determine the emotional tone of text. It automatically classifies text as Positive, Negative, or Neutral. Organizations use sentiment analysis to understand customer opinions from reviews, social media posts, surveys, and feedback.

The main goal is to extract subjective information from text and measure how people feel about a product, service, event, or topic. Sentiment analysis helps businesses make data-driven decisions by analyzing large volumes of textual data quickly and accurately.

VADER Sentiment Analyzer

VADER (Valence Aware Dictionary and Sentiment Reasoner) is a lexicon and rule-based sentiment analysis tool specifically designed for social media content, customer reviews, and short texts. It produces four scores: Positive, Negative, Neutral, and Compound. The Compound score summarizes the overall sentiment of the text on a scale from -1 to +1.

Applications of Sentiment Analysis

- Product review analysis
- Social media monitoring
- Customer feedback evaluation
- Brand reputation management
- Market research and business intelligence
- Political and public opinion analysis

Advantages

- Automates opinion mining
- Saves time and effort
- Helps organizations understand customer needs
- Supports data-driven decision making

Limitations

- Difficulty understanding sarcasm and irony
- Context-dependent interpretations
- Reduced accuracy on domain-specific language
- Mixed sentiments can be challenging to classify

Examples

"This product is amazing!" → Positive

"I hate this service." → Negative

"The product is okay." → Neutral

```
!pip install vaderSentiment
import pandas as pd
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer
analyzer = SentimentIntensityAnalyzer()
reviews = [
    "This mobile phone is excellent",
    "The service was very bad",
    "The product is average",
    "I am happy with the purchase",
    "The delivery was delayed and disappointing"]
results = []
for review in reviews:
    score = analyzer.polarity_scores(review)
    compound = score['compound']
    if compound >= 0.05:
        sentiment = "Positive"
    elif compound <= -0.05:
        sentiment = "Negative"
    else:
        sentiment = "Neutral"
    results.append([review, sentiment])
df = pd.DataFrame(results, columns=['Review', 'Sentiment'])
print(df)
```

```
results.append([review, sentiment])
df = pd.DataFrame(results, columns=['Review', 'Sentiment'])
print(df)
```

*** Collecting vaderSentiment
Downloading vaderSentiment-3.3.2-py2.py3-none-any.whl.metadata (572 bytes)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from vaderSentiment) (2.32.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->vaderSentiment) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->vaderSentiment) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->vaderSentiment) (2.2.3)
Requirement already satisfied: certifi<2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->vaderSentiment) (2024.7.4)
Downloading vaderSentiment-3.3.2-py2.py3-none-any.whl (125 kB)
126.0/126.0 kB 3.6 MB/s eta 0:00:00
Installing collected packages: vaderSentiment
Successfully installed vaderSentiment-3.3.2

	Review	Sentiment
0	This mobile phone is excellent	Positive
1	The service was very bad	Negative
2	The product is average	Neutral
3	I am happy with the purchase	Positive
4	The delivery was delayed and disappointing	Negative

Lab 15: Final Capstone

Customer Review Analysis and Product Recommendation System

Objective

To develop a complete data mining solution that integrates data preprocessing, sentiment analysis, classification, clustering, and recommendation techniques to extract meaningful insights from customer review data and support business decision-making.

Real-World Scenario

E-commerce companies receive thousands of customer reviews every day. Manually analyzing these reviews is difficult and time-consuming. Businesses need automated systems that can determine customer sentiment, identify customer groups, and recommend products based on purchasing behavior.

This capstone project combines multiple data mining techniques to build an intelligent analytics system.

Dataset Description

The project uses a customer review dataset containing:

- Review ID
- Customer ID
- Product Name
- Product Category
- Review Text
- Rating
- Purchase Date
- Customer Age
- Customer Location

Dataset Size

- Approximately 10,000 records
- Multiple product categories
- Textual and numerical attributes

Software Environment

Python Libraries

Pandas

NumPy

Scikit-Learn

Matplotlib

Seaborn

NLTK

VADER Sentiment Analyzer

Practical Objectives

After completing this capstone project, students will be able to:

Clean and preprocess raw data.

Handle missing values and duplicates.

Perform sentiment analysis on customer reviews.

Train classification models.

Cluster customers using K-Means.

Generate product recommendations.

Evaluate model performance.

Create business insights from data.

Project Workflow

Step 1: Data Collection

Load the customer review dataset into Python.

Tasks

Import dataset

Inspect records

Verify data types

Step 2: Data Cleaning

Perform preprocessing operations:

Remove duplicates

Handle missing values

Standardize text

Remove irrelevant records

KPI

Data Quality Score

Missing Value Reduction %

Step 3: Sentiment Analysis

Apply VADER Sentiment Analyzer.

Output Classes

Positive

Negative

Neutral

KPI

Percentage of Positive Reviews

Percentage of Negative Reviews

Average Compound Score

Step 4: Classification

Train a classification model using:

Decision Tree

Naïve Bayes

SVM

Evaluation Metrics

Accuracy

Precision

Recall

Step 5: Customer Clustering

Apply K-Means Clustering.

Purpose

Group customers based on:

Purchase Frequency

Ratings

Spending Behavior

KPI

Number of Clusters

Cluster Characteristics

Step 6: Recommendation Analysis

Use association rule mining techniques:

Apriori

FP-Growth

Metrics

Support

Confidence

Lift

Goal

Recommend products frequently purchased together.

Step 7: Visualization

Create visualizations:

Sentiment Distribution

Customer Clusters

Top Products

Association Rules

Results

Sample Findings

72% Positive Reviews

18% Neutral Reviews

10% Negative Reviews

4 Customer Segments Identified

Top Recommended Product Pair:

Mobile Cover → Screen Protector

Business Insights

Customers are generally satisfied with products.

Electronics category generates the highest engagement.

High-spending customers belong mainly to Cluster 2.

Product bundles can increase sales through cross-selling.

End of Manual